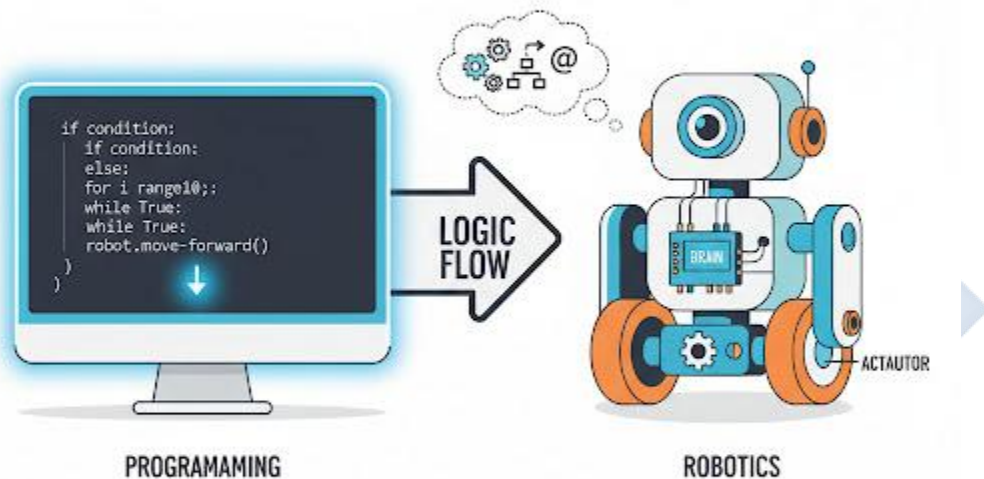


JSS 3 DIGITAL TECHNOLOGIES LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF JSS2 WORK



Content: Quick recap of programming and robotics.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "programming," "algorithm," and "debugging."
2. Recall the difference between block-based (like Scratch) and text-based (like Python) programming.
3. Explain the function of variables, loops, and conditional statements (IF/ELSE).

4. Define "robotics" and list the three main components of a robot (sensors, controllers, actuators).
5. Discuss the relationship between programming (software) and robotics (hardware).

Lesson Content:

1. Review of Core Programming Concepts (JSS2)

- **A. What is Programming?**
 - **Definition:** Programming (or coding) is the process of writing instructions for a computer (or any digital device) to follow to perform a specific task.
 - These instructions are written in a specific "programming language" that the computer understands.
 - **Algorithm:** The *plan* or *step-by-step logic* you create *before* you start coding. It's the "recipe" for solving a problem.
 - **Debugging:** The process of finding and fixing errors ("bugs") in your code.
- **B. Types of Programming Languages (Recap):**
 1. **Block-Based (Visual):** Uses interlocking graphical blocks to build code, like digital LEGOs. It's excellent for beginners.
 - **Example: Scratch.** You drag blocks like "Move 10 steps" or "When green flag clicked."
 2. **Text-Based:** Uses written commands and syntax (grammar rules). This is how all professional software is built.
 - **Example: Python, JavaScript, Java.** (e.g., `print("Hello World")`).
- **C. The Building Blocks of Programming:**
 0. **Variables:** "Boxes" in the computer's memory used to store and label information that can change (e.g., `score = 0`, `playerName = "Aisha"`).
 1. **Conditional Statements (IF/ELSE):** Used for making decisions. They control the flow of the program.
 - **Logic:** IF a condition is true, THEN do this. ELSE (otherwise), do that.
 - **Example:** IF `score > 10`, THEN print "You win!".

2. **Loops (Iteration):** Used to repeat a block of code multiple times to avoid manual repetition.

- **Example:** Instead of writing "Move 10 steps" 100 times, you use a loop:
REPEAT 100 times [Move 10 steps].

2. Review of Robotics Concepts (JSS2)

- **A. What is Robotics?**

- **Definition:** A field of science and engineering that involves the design, construction, operation, and application of **robots**.
- A **robot** is a programmable machine built to automatically carry out a series of complex actions.

- **B. Main Components of a Robot:**

1. **Controller (The "Brain"):** The computer (microprocessor) that processes information and makes decisions. It runs the program (the software).
2. **Sensors (The "Senses"):** Devices that gather information about the robot's environment.
 - *Examples:* Light sensors (to see), touch sensors (to feel), ultrasonic sensors (to detect distance), cameras.
3. **Actuators (The "Muscles"):** The parts that make the robot *move* or *do* something.
 - *Examples:* Motors (to turn wheels), grippers (to pick things up), speakers (to make sound), lights.

- **C. The Link: Programming and Robotics**

- A robot's body (sensors, actuators, wires) is the **hardware**.
- The program (the code) is the **software**.
- **Robotics** is the combination of both. The *programming* gives the "brain" (controller) the instructions to read the "senses" (sensors) and control the "muscles" (actuators).
- *Example:* IF the Touch Sensor is pressed, THEN tell the Motor to stop.

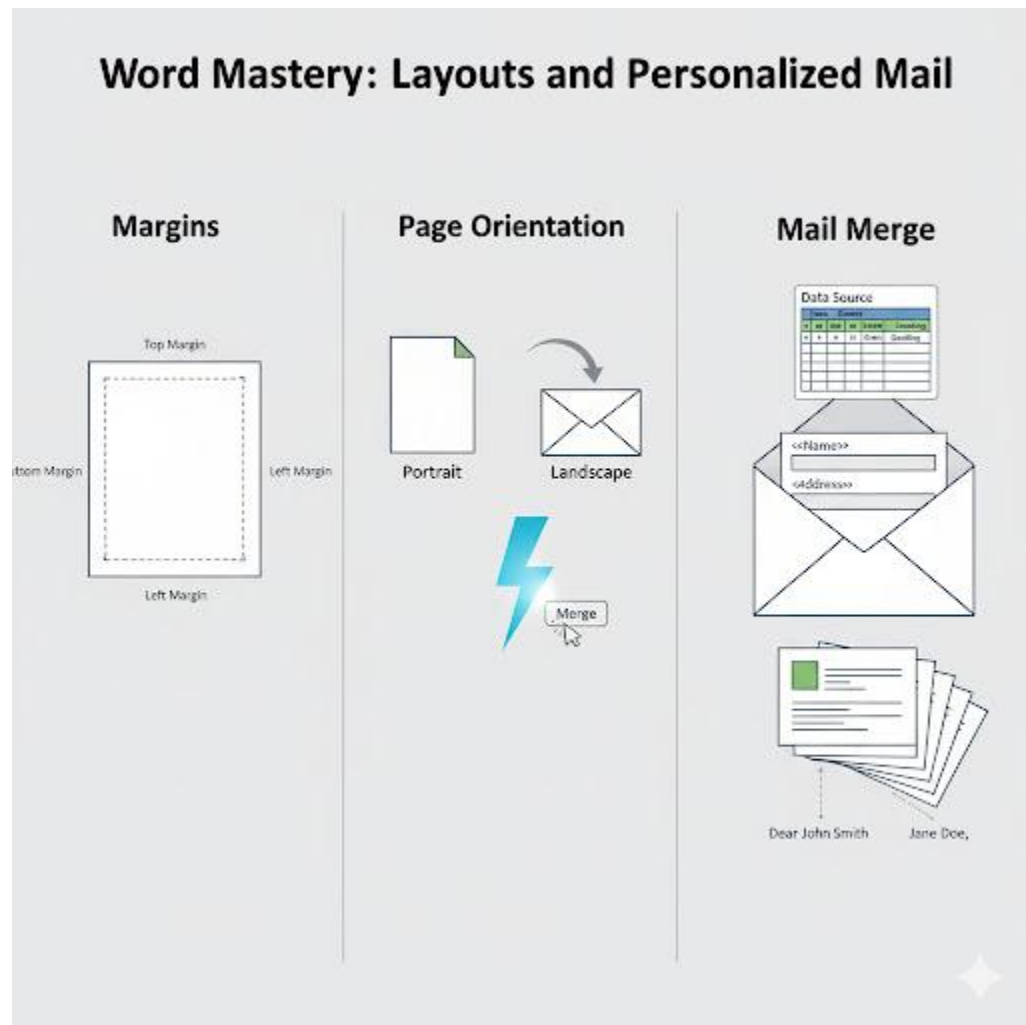
Evaluation:

1. What is an "algorithm"?
2. What is the difference between block-based and text-based programming?
3. What is a "variable" used for in programming?
4. What are the three main components of a robot?
5. What is the "brain" of a robot called?

Assignment:

1. Define Programming, Debugging, and Robotics.
2. In your own words, explain the difference between a loop and a conditional statement.
3. List two (2) examples of sensors and two (2) examples of actuators.
4. Explain the relationship between hardware and software in a robot.
5. Write a simple *pseudocode* (plain English algorithm) for a robot that is designed to follow a black line. (e.g., *Start. Look at the ground with light sensor. IF sensor sees black, move forward. ELSE, turn left.*)

WEEK 2: ADVANCED MS WORD



Content: (I) Page layout (II) Tables (III) Mail merge.

Performance Objectives: By the end of this lesson, students should be able to:

1. Demonstrate how to change page orientation, margins, and size.
2. Create, format, and style a table (e.g., borders, shading, merging cells).
3. Define "Mail Merge" and state its purpose.
4. Perform a simple mail merge to create personalized letters using a data source.
5. Insert headers, footers, and page numbers into a document.

Lesson Content:

1. Page Layout (Formatting Your Document) The "Layout" tab in MS Word controls the look and feel of your entire page before you even type.

- **A. Margins:**

- **Definition:** The empty, white space around the edges (top, bottom, left, right) of your document.
- **Use:** Standard margin is 1 inch (or 2.54 cm). You can make them "Narrow" to fit more text on a page or "Wide" for a more formal look.
- **How:** Layout Tab -> Margins -> Choose a preset (Normal, Narrow, etc.).

- **B. Orientation:**

- **Definition:** The direction your page is facing.
- 2. **Portrait:** Taller than it is wide (like a standard letter). This is the default.
- 3. **Landscape:** Wider than it is tall (like a certificate or a wide chart).
- **How:** Layout Tab -> Orientation -> Choose Portrait or Landscape.

- **C. Page Size:**

- **Definition:** The physical size of the paper you will print on.
- **Common Sizes:**
 - **A4:** The standard size for most of the world (including Nigeria).
 - **Letter:** The standard size for the USA.
- **How:** Layout Tab -> Size -> Choose A4.

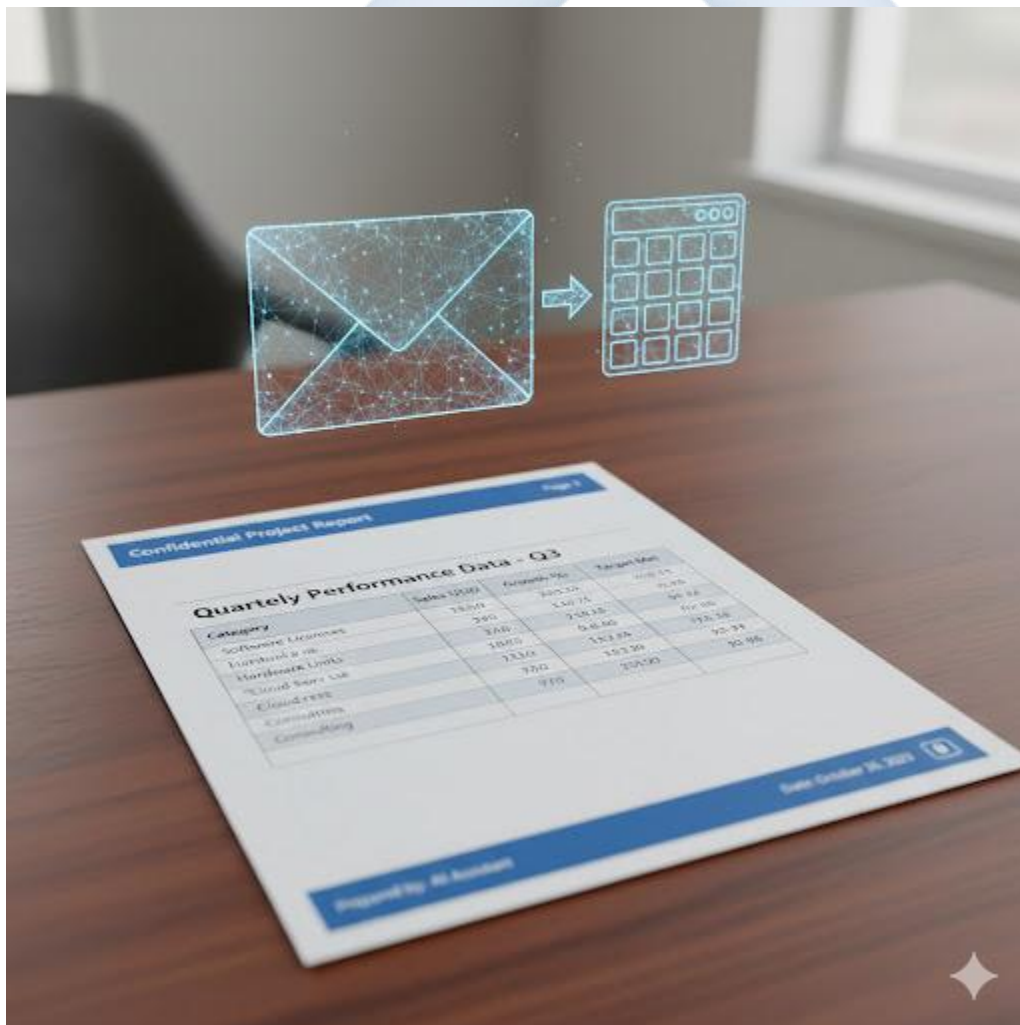
- **D. Headers and Footers:**

- **Definition:** A "header" is text that appears at the *top* of *every* page. A "footer" is text that appears at the *bottom* of *every* page.
- **Use:** For page numbers, document titles, or author's name.
- **How:** Insert Tab -> Header or Footer. (To insert page numbers: Insert Tab -> Page Number).

2. Working with Tables Tables are used to organize data into rows and columns.

- **A. Creating a Table:**

- **How:** Insert Tab -> Table -> Drag your mouse over the grid to select the number of rows and columns.
- **B. Formatting a Table:**
 - When you click on a table, two new tabs appear: **"Table Design"** and **"Layout"**.
 - **Table Design Tab:**
 - **Borders:** Add or remove lines.
 - **Shading:** Change the background color of a cell.
 - **Table Styles:** Choose a pre-formatted, professional look.
 - **Layout Tab (Table):**
 - **Merge Cells:** Combine two or more cells into one large cell.
 - **Split Cells:** Divide one cell into multiple smaller cells.
 - **Insert/Delete:** Add or remove rows and columns.



3. Mail Merge

- **Definition:** Mail Merge is a powerful tool used to create a large number of *personalized* documents from a *single template* and a *data source*.
- **Purpose:** To send the *same* letter to *many different* people, but with each person's unique details (like their name and address) inserted automatically.
- **Example Use:** A school sending a "report card" letter to 1,000 parents. The letter (template) is the same, but the student's name, grade, and parent's name (data) are different for each one.
- **The Two Components:**
 1. **The Template (Main Document):** The MS Word file that contains the main text that will be the same for everyone (e.g., "Dear [Name], your result is...").
 2. **The Data Source:** The file that contains the *list* of information that will change. This is usually an **Excel spreadsheet** or an **MS Access** database (e.g., a list of names, addresses, grades).
- **Simple Steps for Mail Merge:**
 1. **Start:** Go to the Mailings Tab -> Start Mail Merge -> Letters.
 2. **Write:** Type your template letter in Word. (e.g., "Dear...").
 3. **Select Recipients:** Mailings Tab -> Select Recipients -> Use an Existing List (and find your Excel file).
 4. **Insert Merge Field:** Click where you want the name to go, then Mailings Tab -> Insert Merge Field -> Choose First_Name. Your letter will now look like: "Dear <<First_Name>>,"
 5. **Preview:** Mailings Tab -> Preview Results. You will see the <<First_Name>> replaced with the actual names from your list.
 6. **Finish:** Mailings Tab -> Finish & Merge -> Edit Individual Documents (to create one big file) or Print Documents.

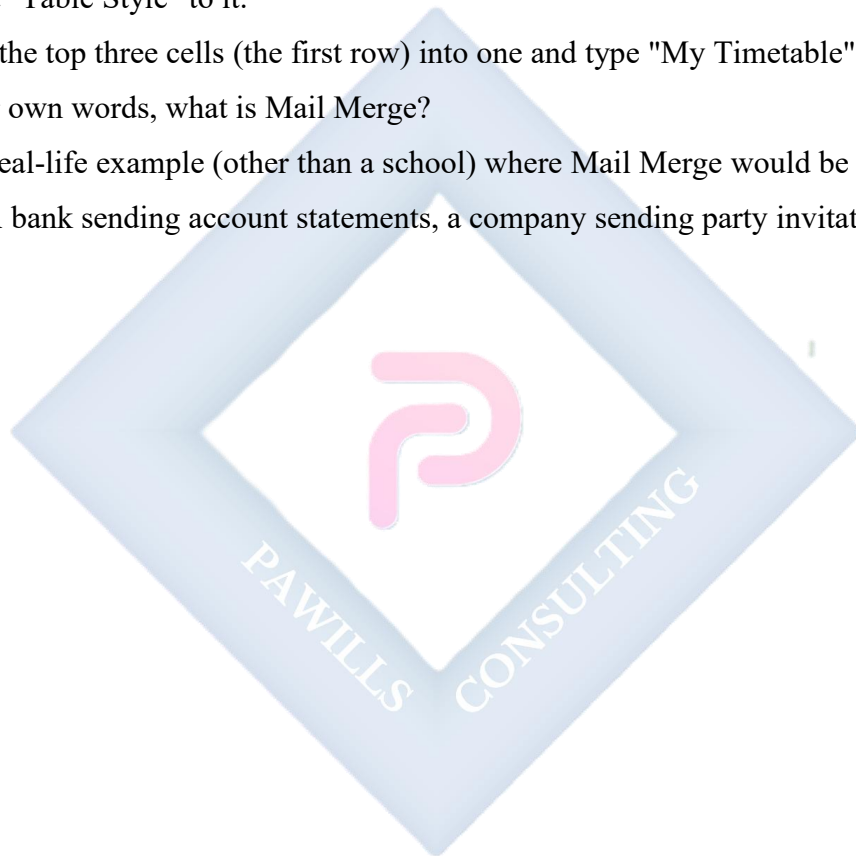
Evaluation:

1. What is the difference between "Portrait" and "Landscape" orientation?
2. What is the difference between a "Header" and a "Footer"?

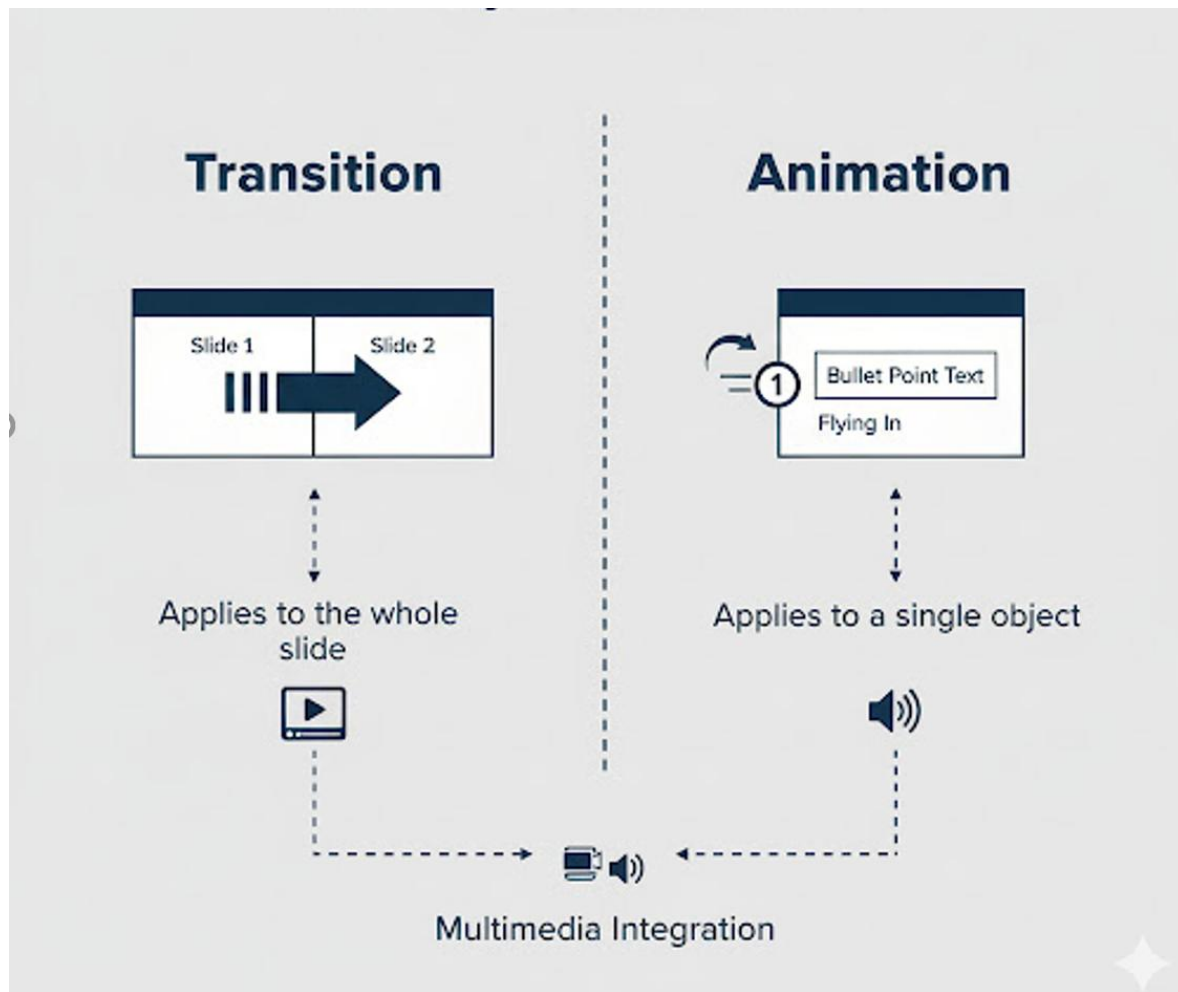
3. How do you combine two table cells into one? (Merge Cells).
4. What is the purpose of Mail Merge?
5. What are the two (2) files needed to perform a Mail Merge?

Assignment:

1. Define Margins, Orientation, and Header.
2. Open MS Word on a computer (if available). Create a 3x4 table (3 columns, 4 rows) and apply a "Table Style" to it.
3. Merge the top three cells (the first row) into one and type "My Timetable" in it.
4. In your own words, what is Mail Merge?
5. List a real-life example (other than a school) where Mail Merge would be very useful.
(e.g., A bank sending account statements, a company sending party invitations to staff).



WEEK 3: ADVANCED MS POWERPOINT



Content: (I) Animation (II) Transitions (III) Multimedia integration.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Slide Transition" and apply one to a slide.
2. Define "Animation" and apply an effect to an object (text or image).
3. Differentiate clearly between a Transition and an Animation.
4. Insert multimedia (Video and Audio) into a presentation.
5. Explain the purpose of "Trigger" in animations.

Lesson Content:

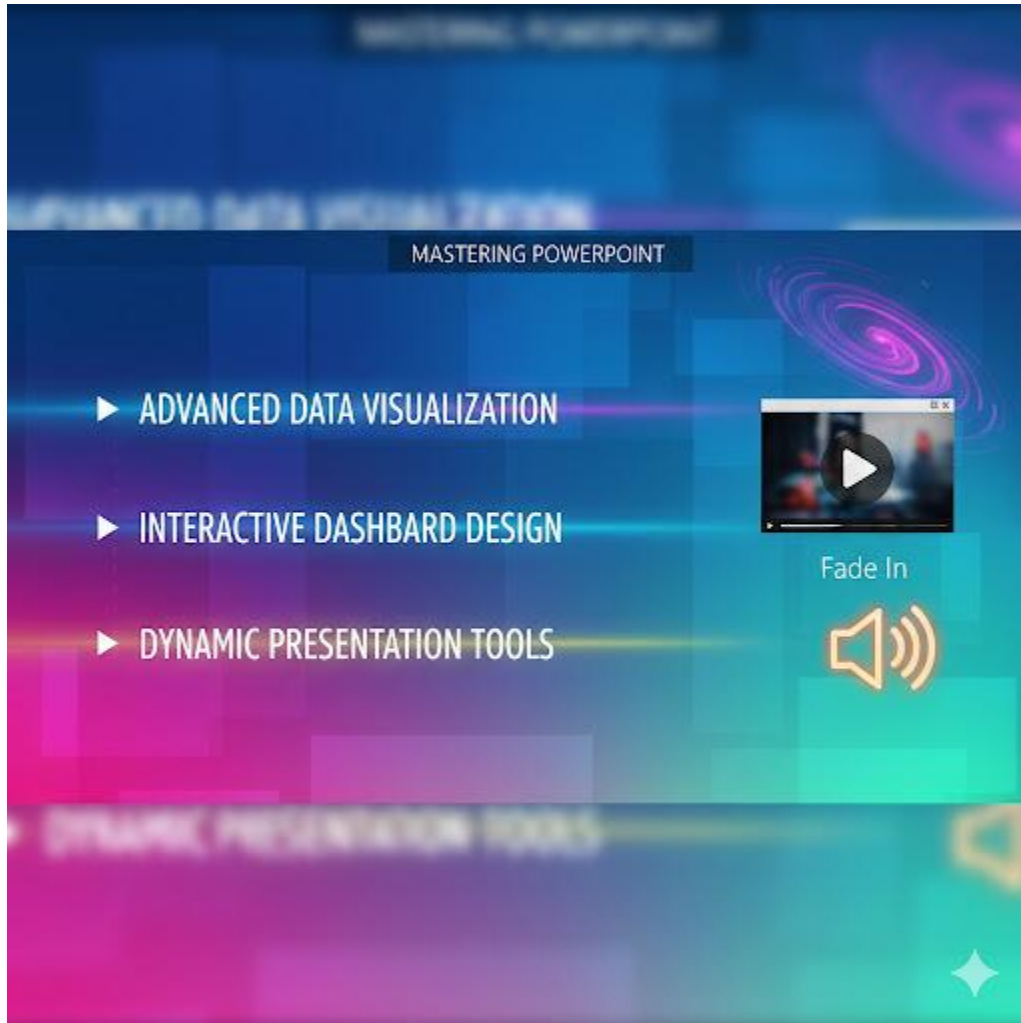
1. Slide Transitions

- **Definition:** A **Transition** is the *visual effect* that occurs when you **move from one slide to the next** during a presentation.
- **It applies to the *entire slide*.**
- **Purpose:** To make the presentation flow smoothly and professionally, instead of just "cutting" abruptly from one slide to the next.
- **Examples:**
 - Fade: The old slide fades out and the new one fades in.
 - Wipe: The new slide "wipes" across the old one.
 - Push: The new slide "pushes" the old one off the screen.
- **How to Apply:**
 1. Click on the slide you want to *transition to*.
 2. Go to the **Transitions Tab**.
 3. Choose an effect (e.g., Morph, Fade, Push).
 4. You can set the Duration (how fast it is) and click Apply to All to make all slides the same.

2. Animations

- **Definition:** An **Animation** is a *visual effect* applied to a **single object on a slide**.
- **It applies to *individual items*** (like a piece of text, a picture, or a shape).
- **Purpose:** To control the flow of information. It allows you to reveal points one by one, keeping the audience's attention.
- **Types of Animations:**
 1. **Entrance:** How the object *enters* the slide (e.g., Fly In, Appear, Zoom).
 2. **Emphasis:** An effect to draw attention to an object *already on* the slide (e.g., Spin, Pulse).
 3. **Exit:** How the object *leaves* the slide (e.g., Fly Out).
- **How to Apply:**
 1. Click on the *object* you want to animate (e.g., a text box).

2. Go to the **Animations Tab**.
3. Choose an effect (e.g., Fly In).
4. Use the "**Animation Pane**" to see the list of your animations and change their order.



3. Transition vs. Animation (The Key Difference)

Feature	Slide Transition	Animation
What it applies to?	The entire slide.	A single object (text, image) on the slide.
When it happens?	Between slides.	On a slide (when you click).
Tab in PowerPoint?	Transitions Tab	Animations Tab

4. Multimedia Integration "Multimedia" means using more than one type of media. This makes presentations dynamic and engaging.

- **A. Inserting Video:**

- **How:** Insert Tab -> Video -> This Device (to add a video file saved on your computer).
- The video will be placed on the slide, and you can play it during your presentation.

- **B. Inserting Audio:**

- **How:** Insert Tab -> Audio -> Audio on My PC (to add a saved music or sound file) or Record Audio (to record your own voice).
- **Use:** You can add background music to a slide or record narration to explain a concept.

- **C. Triggers:**

- **Definition:** A "trigger" is an advanced animation tool. It starts an animation *only* when you click a *different, specific* object.
- **Example:** You have a picture of a car. You can set a "trigger" so that when you click the picture of the car, an *audio* file of a "vroom" sound plays.
- **How:** Animations Tab -> Trigger -> On Click of...

Evaluation:

1. What is a "Slide Transition"?
2. What is an "Animation"?
3. What is the main difference between a transition and an animation?
4. What are the three main types of animation effects? (Entrance, Emphasis, Exit).
5. How do you insert a video file into your presentation?

Assignment:

1. Open PowerPoint (if available) and create 3 slides.
2. Apply a *different* "Transition" to Slide 2 and Slide 3.
3. On Slide 2, add a title and a list of three (3) items. Apply an "Entrance" animation (like Fly In) to each of the three items so they appear one by one.
4. In your own words, explain what "Multimedia Integration" means.
5. What is the purpose of the "Animation Pane"?

WEEK 4: INTRODUCTION TO DIGITAL MEDIA



DIGITAL MEDIA

Content: (I) Meaning and importance (II) Creating and editing digital content.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Media" and "Digital Media."
2. Differentiate between "Digital Media" and "Traditional Media."
3. List the five (5) main categories of digital media.
4. State the importance of digital media in modern life (e.g., communication, education).
5. Identify common tools used for creating and editing digital content.

Lesson Content:

1. What is Media?

- **Media:** The plural of "medium." It refers to the *channels* or *tools* used to store and deliver information or data.
- **Traditional Media (or "Analog Media"):** These are the "old" forms of media that are static and one-way.

- *Examples:* Printed books, newspapers, magazines, physical photographs, radio, television.

2. What is Digital Media?

- **Definition:** Digital Media is any form of media content that is created, viewed, distributed, and stored in a **digital (electronic) format**.
- It is "digital" because it is based on code (binary 0s and 1s) that a computer can read.
- **Key Feature:** Digital media is often **interactive, dynamic**, and can be easily **shared** and **modified**.

3. Categories of Digital Media

Digital media is not just one thing. It includes:

1. **Digital Text:** (e.g., A web page, an e-book, a blog post, a WhatsApp message).
2. **Digital Images:** (e.g., Photos from your phone, JPEGs, PNGs, GIFs, digital paintings).
3. **Digital Audio:** (e.g., MP3 files, podcasts, streaming music, voice notes).
4. **Digital Video:** (e.g., Movies on Netflix, YouTube videos, TikTok videos, video calls).
5. **Interactive Media:** (e.g., Video games, websites, mobile apps, virtual reality).

4. Importance of Digital Media

Digital media has revolutionized every aspect of modern life:

- **Communication:** Allows for instant, global, and multimedia communication (e.g., email, social media, video conferencing).
- **Education:** Provides access to unlimited information and learning resources (e.g., e-books, online courses, educational videos on YouTube).
- **Business (Marketing):** Allows businesses to reach customers around the world through social media, websites, and digital ads.
- **Entertainment:** Provides "on-demand" entertainment (e.g., Netflix, Spotify).
- **Information (News):** News is now instant and global, delivered through news websites and social media platforms.



5. Creating and Editing Digital Content

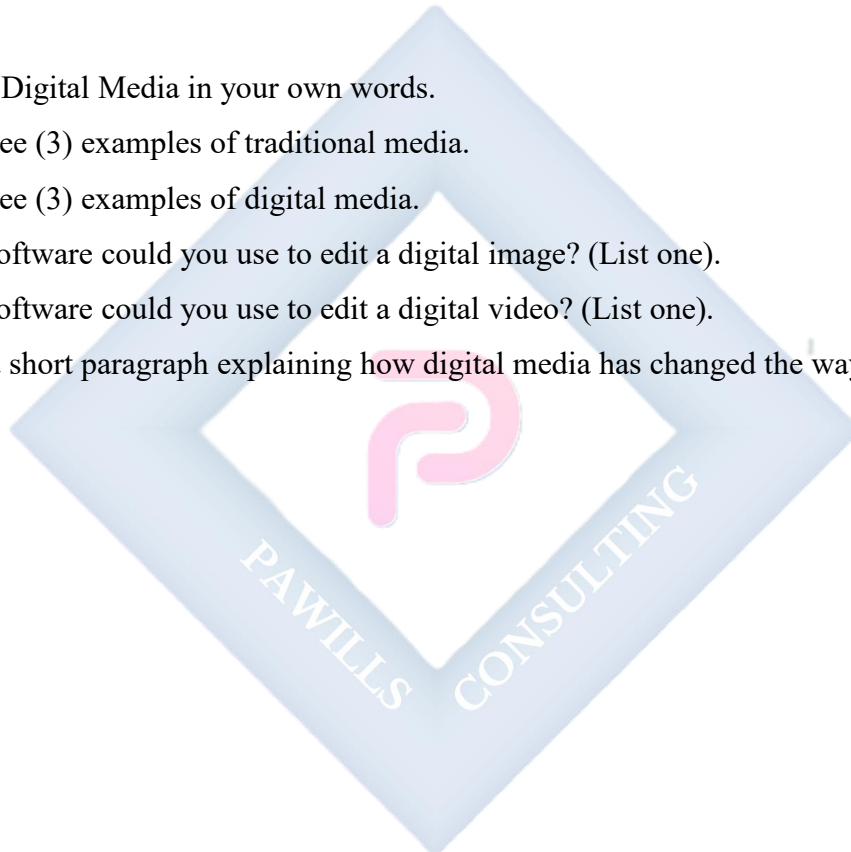
- **"Creating"** is the act of making new digital media (e.g., taking a photo, recording your voice, typing a blog post).
- **"Editing"** is the act of modifying, improving, or combining existing media.
- **Common Tools:**
 - **Text:** MS Word, Google Docs (Creating/Editing).
 - **Images:** Adobe Photoshop (Professional), **Canva** (Beginner/Online), phone's photo editor (Editing).
 - **Audio:** Audacity (Free software), phone's voice recorder (Creating).
 - **Video:** Adobe Premiere Pro (Professional), CapCut (Mobile/Online), iMovie (Apple) (Editing).

Evaluation:

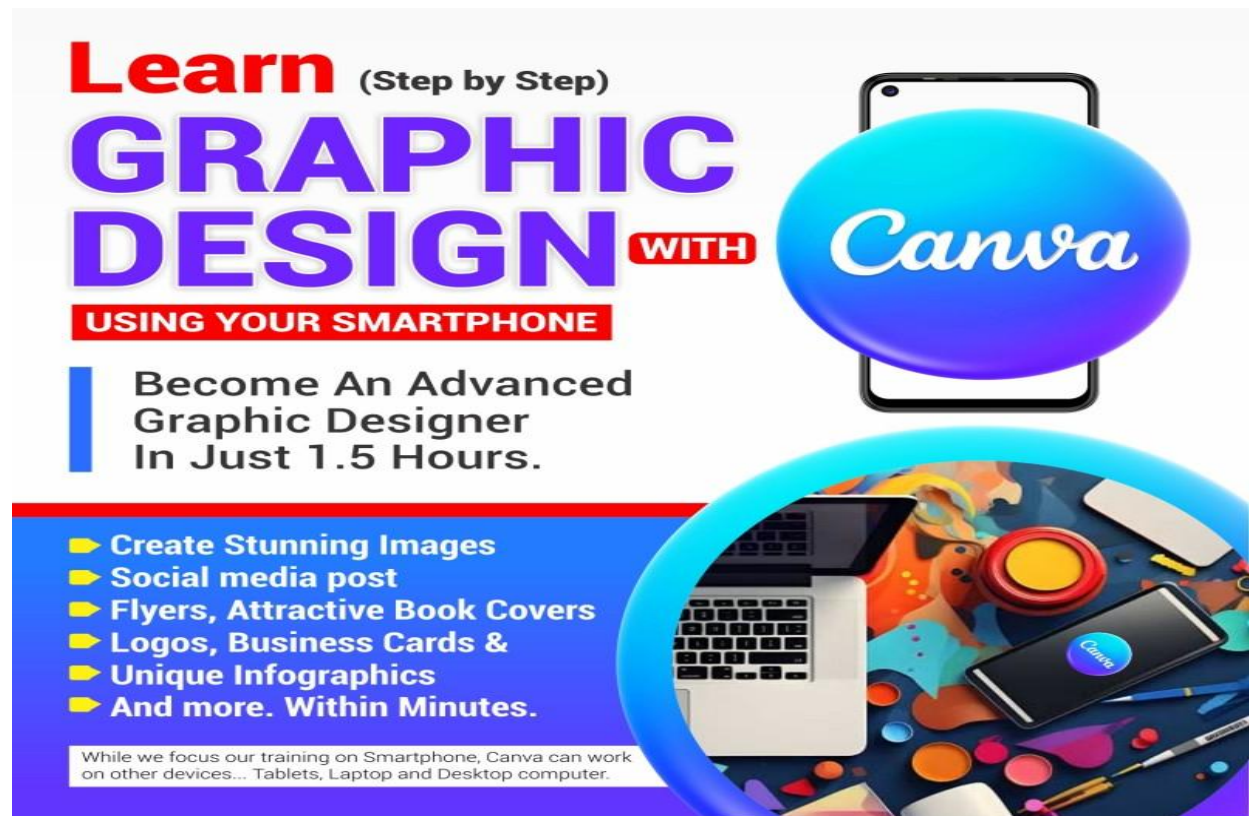
1. What is the difference between "Traditional Media" and "Digital Media"?
2. List the five (5) main categories of digital media.
3. Give one example of "Digital Audio" and one of "Interactive Media."
4. List three (3) reasons why digital media is important.
5. What is the difference between "creating" and "editing" digital content?

Assignment:

1. Define Digital Media in your own words.
2. List three (3) examples of traditional media.
3. List three (3) examples of digital media.
4. What software could you use to edit a digital image? (List one).
5. What software could you use to edit a digital video? (List one).
6. Write a short paragraph explaining how digital media has changed the way *you* learn in school.



WEEK 6: GRAPHIC DESIGN WITH CANVA



Learn (Step by Step)
GRAPHIC DESIGN **WITH**
USING YOUR SMARTPHONE

Become An Advanced Graphic Designer In Just 1.5 Hours.

- ▶ Create Stunning Images
- ▶ Social media post
- ▶ Flyers, Attractive Book Covers
- ▶ Logos, Business Cards &
- ▶ Unique Infographics
- ▶ And more. Within Minutes.

While we focus our training on Smartphone, Canva can work on other devices... Tablets, Laptop and Desktop computer.

The image is a promotional graphic for Canva's graphic design training. It features a blue and purple color scheme. At the top, the word 'Learn' is in red, followed by '(Step by Step)' in black. Below this, 'GRAPHIC DESIGN' is in large, bold, blue letters, with 'WITH' in a red box. Underneath, 'USING YOUR SMARTPHONE' is in a red box. To the right, there is a smartphone icon with a blue oval containing the 'Canva' logo. Below the main text, a blue box contains a list of design tasks. At the bottom, a small text box states that Canva works on various devices. The background includes a circular inset showing a laptop, a smartphone, and various design tools like paint palettes and brushes.

Content: (I) Designing posters (II) flyers (III) social media posts.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Graphic Design" and "Canva."
2. Log in to Canva and navigate its basic interface.
3. Select an appropriate template for a specific task (e.g., a poster).
4. Modify a template by changing text, fonts, and colors.
5. Add and edit elements (like shapes and photos) and download the final design.

Lesson Content:

1. What is Graphic Design?

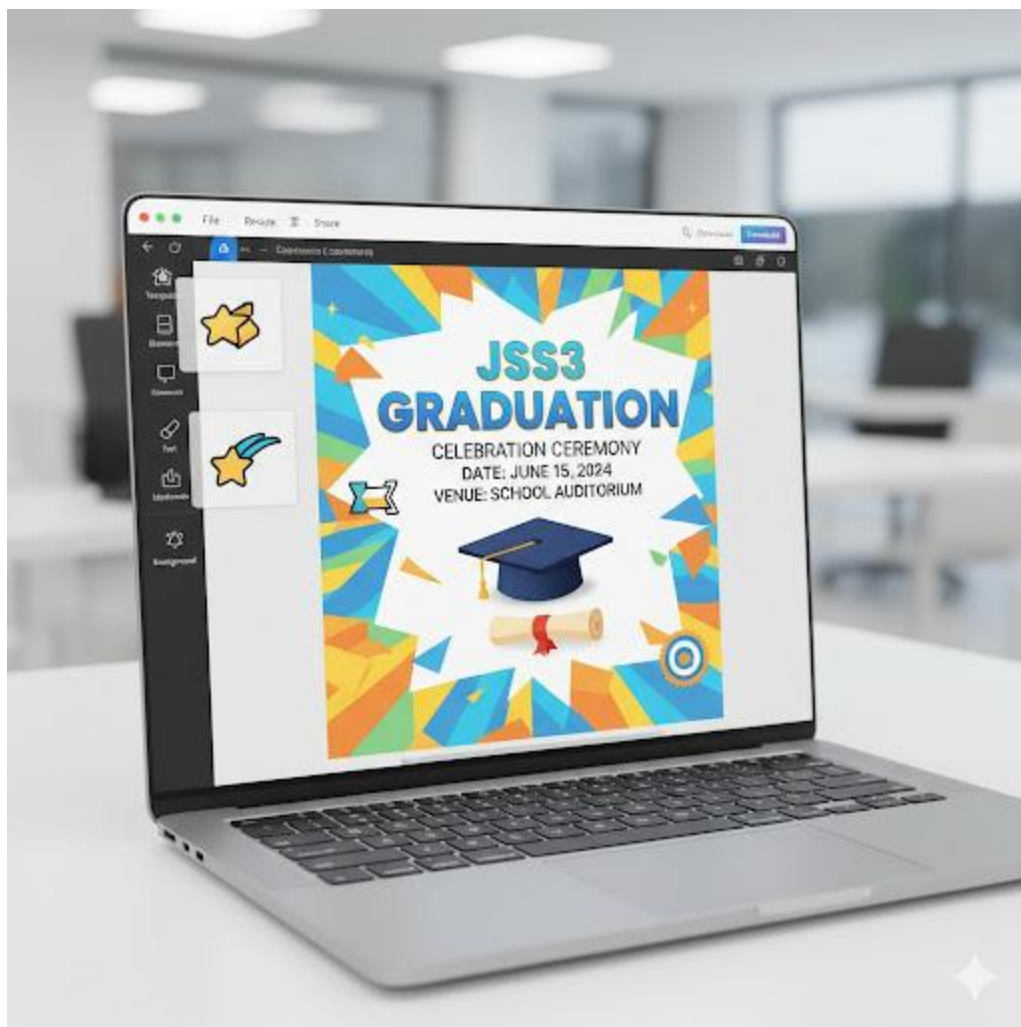
- **Definition:** Graphic Design is the art of **visual communication**. It is the process of combining **text, images, and colors** to create a visual message that informs, persuades, or engages an audience.
- **Examples:** Company logos, posters, book covers, website layouts, T-shirt designs.

2. What is Canva?

- **Definition:** Canva is a powerful and popular **online (web-based) graphic design tool**.
- **Why it's popular:** It is designed for beginners. It makes design easy by providing thousands of pre-made **templates** (layouts) that users can edit.
- **Access:** It is used through a web browser (www.canva.com) or as a mobile app. It is "cloud-based," meaning your designs are saved online.

3. Navigating the Canva Interface

- **Homepage:** Shows your recent designs and a search bar ("What will you design today?").
- **Templates:** The "library" of pre-made designs. You can search for "Poster," "Instagram Post," "Flyer," etc.
- **Editor:** The main workspace where you create your design.
- **The Editor's Side Panel:**
 - **Design/Templates:** To choose a layout.
 - **Elements:** To add shapes, lines, icons, photos, and videos.
 - **Text:** To add headings and body text.
 - **Uploads:** To upload your own pictures (e.g., a school logo).



4. Practical: How to Design a School Poster

- **Objective:** Design a poster for a "JSS3 Graduation Party."
- **Step-by-Step Guide:**
 1. **Log In:** Go to canva.com and log in (e.g., with a Google account).
 2. **Choose a Template:** In the search bar, type "School Party Poster" or "Event Poster." Browse the results and click on one you like.
 3. **Edit the Text:**
 - The template will have "dummy" text. Double-click the text to change it.
 - Change the main heading to "JSS3 GRADUATION PARTY!"
 - Change the other text to include the **Date, Time, and Venue** (e.g., "School Hall").

- You can change the **Font** and **Font Size** from the toolbar at the top.

4. **Change Colors:**

- Click on an element (like the background). A color box will appear on the top toolbar. Click it to choose a new color.
- **Tip:** Use your school's colors (e.g., green and white).

5. **Add Elements:**

- Go to the Elements tab on the left.
- Search for "graduation cap" or "balloons" and drag the icon onto your poster.
- You can resize and move it.

6. **Download:**

- When finished, click the **Share** button (top right).
- Click **Download**.
- Choose the file type: **PNG** (best for quality) or **JPG** (small file size).
- Click Download.

5. **Designing Flyers and Social Media Posts**

- The process is the *exact same*, but you start with a different template size.
- **Flyer:** A standard A4 or A5 paper, designed to be printed and handed out.
- **Social Media Post (e.g., Instagram Post):** A perfect square, designed to look good on a phone screen.

Evaluation:

1. What is "Graphic Design"?
2. What is "Canva"?
3. What is a "template" in Canva?
4. What is the difference between the "Text" tab and the "Elements" tab in the Canva editor?
5. What are the two best file types to download your design as? (PNG/JPG).

Assignment (Practical):

1. On a computer or phone, go to Canva.com and sign up for a free account.
2. Search for "School Poster" or "Class Timetable" in the templates.
3. Create a **Class Timetable** for JSS3.
4. Customize it: Change the title, change the colors, and add at least one "Element" (like a book icon or a pencil).
5. (If possible) Download your design as a PNG or JPG and share it with your teacher.



WEEK 8: PHOTO EDITING AND VISUAL COMMUNICATION



Content: (I) Using online tools for design and branding (II) Visual communication.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Visual Communication."
2. Define "Photo Editing" and its basic functions.
3. Identify basic photo editing tools (e.g., crop, filter, brightness, contrast).
4. Define "Branding" and the role of visual elements (logo, color).
5. Explain how online tools (like Canva) are used for branding.

Lesson Content:

1. What is Visual Communication?

- **Definition:** The practice of using **visual elements** (images, colors, text, charts, drawings) to communicate a message, idea, or feeling.
- **Principle:** "A picture is worth a thousand words." Visuals are often faster, more emotional, and more universal than text alone.
- **Examples:** A traffic light (red means "stop"), a company's logo, a poster, the icons on your phone.

2. Photo Editing

- **Definition:** The process of *altering* or *improving* a digital image. This is a key part of visual communication.
- **Basic Photo Editing Tools (Found in Canva, phone editors, etc.):**
 1. **Crop:** To cut out unwanted parts of an image.
 2. **Rotate/Flip:** To change the orientation.
 3. **Filters:** Pre-set "looks" that change the color and mood of a photo (e.g., "Black & White").
 4. **Adjust (Manual Editing):**
 - **Brightness:** Makes the photo lighter or darker.
 - **Contrast:** The difference between the light and dark areas. (High contrast = sharp and "punchy").
 - **Saturation:** The *intensity* of the colors. (Low saturation = grey/faded, High saturation = very colorful).
 - **Sharpness:** Makes the edges of objects clearer.

3. What is Branding?

- **Definition:** Branding is the *identity* of a company, product, or person. It's the "feeling" or "personality" that people associate with you.
- **Visual Identity (Branding):** This is the *visual* part of your brand. It's how people *recognize* you.
- **Key Visual Elements of a Brand:**

1. **Logo:** The main symbol that represents the brand (e.g., the Nike "swoosh," the "M" for McDonald's).
2. **Color Palette:** A set of specific, consistent colors (e.g., MTN is always Yellow; Coca-Cola is always Red).
3. **Typography (Fonts):** The specific fonts used (e.g., Coca-Cola's "script" font).

4. Using Online Tools (like Canva) for Branding Canva is not just for one-time posters; it's a powerful *branding* tool.

- **Brand Consistency:** To build a brand, you must be **consistent**. Your logo, colors, and fonts should be the *same* on your poster, your website, and your social media.
- **How to use Canva for Branding:**
 1. **Create a Logo:** Use Canva's Logo templates.
 2. **Define Colors:** Choose 2-3 "brand colors" and write down their *hex codes* (e.g., #FF0000 for pure red). Use these same colors for *all* your designs.
 3. **Create Templates:** Design a "master" social media post. When you need to make a new post, you *duplicate* this template and just change the text. This keeps the *look* consistent.
- **Other Tools:** Many online photo editors (like Fotor, Pixlr) or mobile apps (like Snapseed) can be used to apply filters and adjustments to make your photos look professional and consistent.

Evaluation:

1. What is "Visual Communication"?
2. What is "photo editing"?
3. What does the "Crop" tool do?
4. What is "Branding"?
5. What are the three (3) main visual elements of a brand? (Logo, Colors, Fonts).

Assignment:

1. Define Visual Communication and Branding.

2. What is the difference between "Brightness" and "Saturation" in photo editing?
3. Why is "consistency" (using the same colors and logo) important in branding?
4. Find a popular Nigerian brand (e.g., MTN, Glo, Dangote). What are its main "brand colors"?
5. Explain how a good brand logo (like the Nike swoosh) is an example of effective visual communication.



WEEK 9: COPYRIGHT AND MEDIA ETHICS



Content: (I) Digital rights (II) Fair use policies.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Intellectual Property" and "Copyright."
2. Differentiate between "Copyright" and "Plagiarism."
3. Define "Fair Use" (or Fair Dealing).
4. List two (2) examples of Fair Use.
5. Define "Creative Commons" as a type of license.

Lesson Content:

1. What is Intellectual Property (IP)?

- **Definition:** "Intellectual Property" refers to creations of the **mind**. It is "property" (like a car or a house), but it is *intangible* (you can't touch it).
- **Examples:** A story you write, a song you compose, a painting you make, a logo you design, an invention.

2. What is Copyright?

- **Definition:** Copyright is a **legal right** that *automatically* gives the creator of an original creative work (like a book, song, or photo) the *exclusive* right to control how it is used, copied, and distributed.
- **What it protects:** Music, books, movies, photos, software, graphic designs.
- **The "©" Symbol:** This is the symbol for copyright. It means "All Rights Reserved." You cannot use this work without *permission* (a license) from the owner.
- **Copyright vs. Plagiarism (JSS2 Recap):**
 - **Plagiarism:** An *academic* offense. It is *claiming* someone else's work is your own (cheating).
 - **Copyright Infringement:** A *legal* offense. It is *using, copying, or selling* someone else's work without permission, even if you give them credit. (e.g., uploading a full Nollywood movie to your YouTube channel).

3. Digital Rights and Media Ethics

- **Digital Rights:** Your rights as a creator and user of digital media. This includes your right to privacy, freedom of speech, and the right to *control* your own creations (copyright).
- **Media Ethics:** The moral principles that guide how we use and create media.
 - **Ethical Rule:** Just because you *can* "right-click and save" an image from Google, doesn't mean you *should* use it in your project without permission or giving credit.

4. "Fair Use" (or "Fair Dealing" in Nigeria)

- **Definition:** "Fair Use" is a *limited exception* to the copyright law. It allows you to use a *small part* of a copyrighted work *without permission* for specific, limited purposes.
- **Purpose:** To balance the rights of the creator with the public's right to information and education.
- **Allowed Uses (Generally):**
 1. **Criticism/Review:** Using a 10-second clip of a movie in your video *review* of that movie.
 2. **News Reporting:** A news channel showing a photo of an event.
 3. **Education/Teaching:** A teacher photocopying one chapter (not the whole book) for students.
 4. **Parody:** Making a "comedy" version of a song.
- **What is NOT Fair Use:**
 - Using the *whole* work.
 - Using it for *commercial* purposes (to make money).
 - Downloading a "cracked" (pirated) version of MS Word.

5. Creative Commons (CC)

- **Definition:** An alternative to "All Rights Reserved." A **Creative Commons (CC)** license is a way for creators to give the public *permission in advance* to share and use their work *for free*, as long as they follow certain rules.
- **The "Some Rights Reserved" License:**
 - **CC-BY (Attribution):** The most common. "You can use my photo for free, even to make money, as long as you *give me credit* (attribute me)."
- **Where to find CC content:** Wikipedia, Flickr, and many online photo sites.

Evaluation:

1. What is "Intellectual Property"?
2. What is "Copyright"?
3. What is the main difference between Copyright Infringement and Plagiarism?
4. What is "Fair Use"?

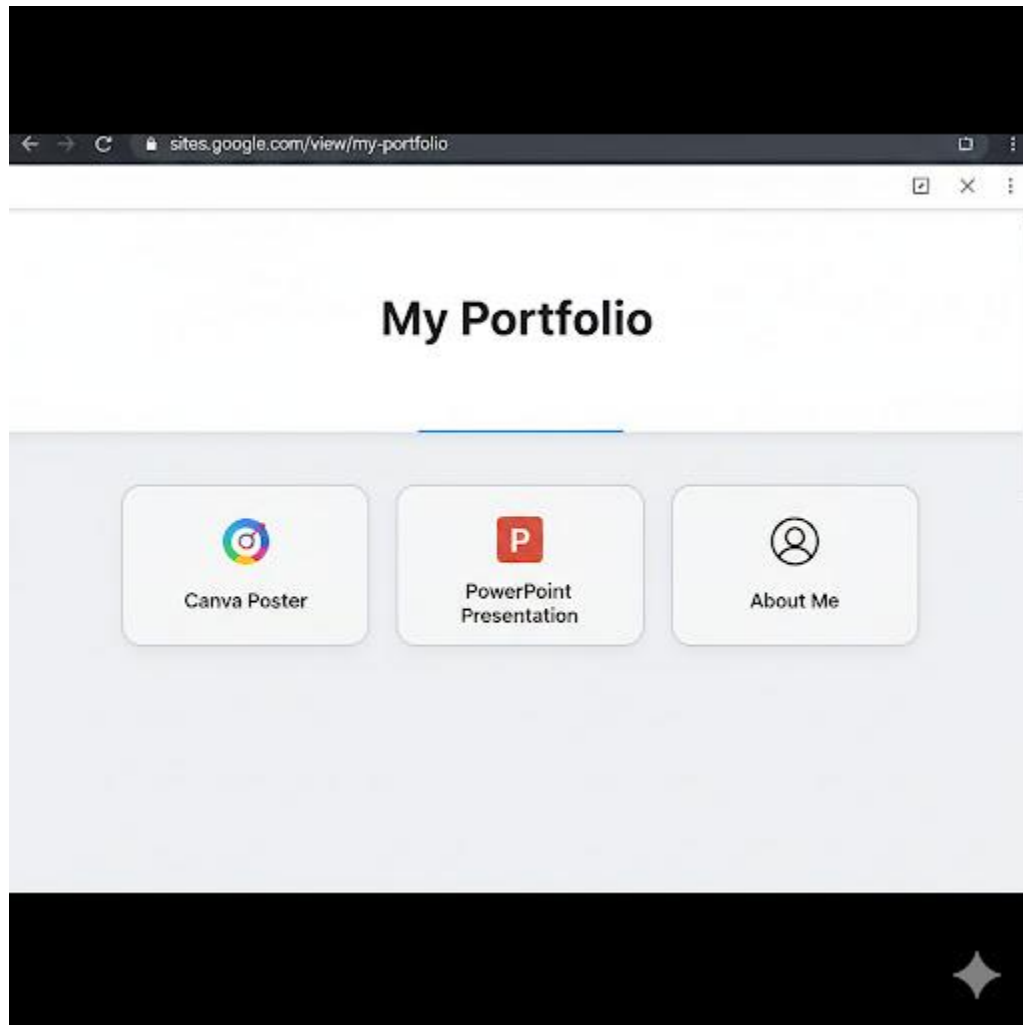
5. What is a "Creative Commons" (CC) license?

Assignment:

1. Define Copyright and Fair Use.
2. If you write a story, who owns the copyright to it?
3. Your friend downloads a new movie from a pirate website instead of paying for it. Are they violating Copyright, Fair Use, or Plagiarism?
4. You are doing a school project on "Lions." You copy a photo of a lion from a Google search and put it on your project cover. Is this allowed? What *should* you do? (e.g., Find a "Creative Commons" photo and give credit).
5. What does the "CC-BY (Attribution)" license mean?



WEEK 10: DIGITAL PORTFOLIO CREATION



Content: (I) Compiling digital works and projects.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Digital Portfolio."
2. State the purpose of a digital portfolio.
3. Identify different platforms for creating a portfolio.
4. List the types of content that can be included in a portfolio.
5. Create a simple, one-page digital portfolio using a tool like Google Sites.

Lesson Content:

1. What is a Digital Portfolio?

- **Definition:** A **digital portfolio** is an *online collection* of your *best work* that showcases your skills, projects, and accomplishments.
- **Old vs. New:** In the past, an artist would carry a large, physical "portfolio" (a big folder) of their drawings. A *digital* portfolio puts this online, making it accessible to anyone, anywhere in the world.
- **It is your *digital resume*.** It *shows* what you can do, rather than just *telling*.

2. Purpose of a Digital Portfolio

- To get a job.
- To apply for admission to a university.
- To get freelance clients.
- To track your own progress and skills.
- To showcase your creativity (e.g., your graphic designs, photos, or writing).

3. What to Include in Your Portfolio (Your "Works")

A portfolio for a JSS3 student could include:

- **Graphic Designs:** Your best posters, flyers, or logos made in Canva (from Week 6).
- **Presentations:** Your best MS PowerPoint presentation (from Week 3).
- **Writing:** A narrative or expository essay you are proud of (from English class).
- **Code:** A link to a Scratch project you built (from JSS2).
- **Photos/Videos:** Any photos or videos you have edited.
- **"About Me" Page:** A short biography introducing yourself, your skills, and your goals.
- **Contact Page:** A way for people to contact you (e.t., a professional email).

4. Platforms for Creating a Digital Portfolio

- **1. Google Sites (Recommended - Easiest):**
 - A *free*, simple, drag-and-drop website builder from Google.

- It connects directly to your Google Drive, so you can easily add your Google Docs, Slides, and images.
- **2. Canva (Also Easy):**
 - Canva has "Website" templates. You can design a beautiful, one-page website and publish it with a few clicks.
- **3. Wix / Weebly:**
 - More advanced website builders that offer more templates and features.
- **4. Behance:**
 - A professional portfolio site specifically for *designers* and *artists*.

5. Practical: How to Create a Simple Portfolio with Google Sites

1. **Go to:** sites.google.com (and log in with your Google account).
2. **Start:** Click the + button to start a "Blank site."
3. **Title:** Give your site a title (e.g., "Aisha's Digital Portfolio").
4. **Add Content:** Use the **Insert** panel on the right.
 - Insert a Text Box to write your "About Me" introduction.
 - Insert an Image (e.g., a professional-looking photo of yourself or a logo you designed).
 - Use the Google Drive button to find and insert your best PowerPoint (Slides) or Word (Docs).
 - Use a Content Block (e.g., the one with a picture and text) to show a poster you made in Canva (upload the PNG) and write a short description of the project.
5. **Publish:**
 - Click the **Publish** button (top right).
 - Choose a web address (e.g., sites.google.com/view/aisha-portfolio).
 - Click Publish. Your portfolio is now "live" on the internet.

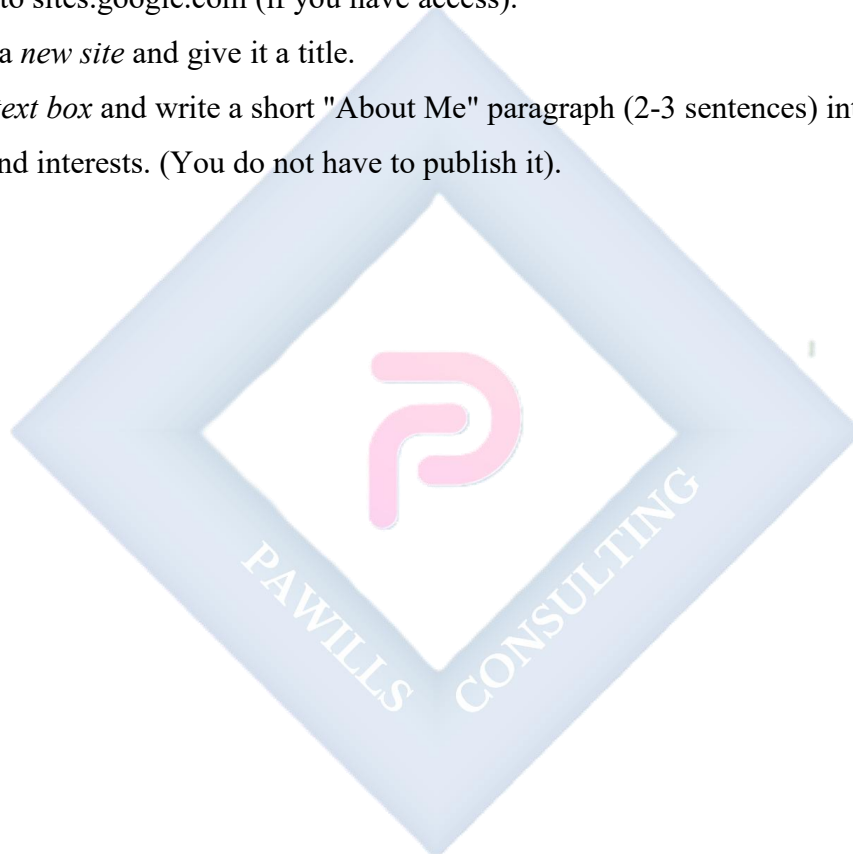
Evaluation:

1. What is a digital portfolio?
2. What is the main *purpose* of creating a portfolio?

3. List three (3) types of "works" or "projects" you could include.
4. What is the difference between a traditional (physical) portfolio and a digital one?
5. Name one (1) free online tool you can use to build a digital portfolio.

Assignment:

1. Define "Digital Portfolio."
2. Why is a portfolio better than a simple CV or resume? (Hint: "Show, don't tell").
3. Log in to sites.google.com (if you have access).
4. Create a *new site* and give it a title.
5. Add a *text box* and write a short "About Me" paragraph (2-3 sentences) introducing your skills and interests. (You do not have to publish it).



WEEK 11: REVISION



Note: This week is dedicated to a comprehensive review of all topics covered in the First Term, especially in preparation for the Basic Education Certificate Examination (BECE).

Revision Checklist:

1. JSS2 Review (Week 1):

- Programming: Algorithm, Debugging, Variables, Loops, Conditionals.
- Robotics: Sensors, Controllers, Actuators.

2. Advanced MS Word (Week 2):

- Page Layout: Margins, Orientation (Portrait/Landscape), Page Size (A4).
- Tables: Create, format, merge/split cells.

- Headers/Footers/Page Numbers.
 - Mail Merge: Purpose (personalized bulk letters), Template, Data Source.
3. **Advanced MS PowerPoint (Week 3):**
- Transition: Effect *between* slides.
 - Animation: Effect *on* an object (text/image). Types: Entrance, Emphasis, Exit.
 - Multimedia: Inserting Video and Audio.
4. **Digital Media (Week 4):**
- Traditional vs. Digital Media.
 - Categories: Text, Image, Audio, Video, Interactive.
 - Importance: Communication, Education, Business.
 - Tools: Canva (Image), CapCut (Video).
5. **Graphic Design (Week 6):**
- Definition: Visual communication (text + images + color).
 - Canva: Online, template-based tool.
 - Process: Choose template, edit text, change colors, add elements, download (PNG/JPG).
6. **Visual Communication & Branding (Week 8):**
- Photo Editing: Crop, Brightness, Contrast, Saturation.
 - Branding: A company's identity.
 - Visual Identity: Logo, Color Palette, Fonts.
 - Consistency: Using the same visuals everywhere.
7. **Ethics and Rights (Week 9):**
- Intellectual Property: Creations of the mind.
 - Copyright (©): Legal right to *control* your work.
 - Plagiarism: *Claiming* others' work is yours (academic cheating).
 - Fair Use: Limited exception for education, review, or criticism.
 - Creative Commons (CC): "Some Rights Reserved" - free to use with rules (like giving credit).
8. **Digital Portfolio (Week 10):**
- Definition: An online collection of your best work.
 - Purpose: To "show" your skills for jobs or university.

- Platforms: Google Sites, Canva.

Practical Revision:

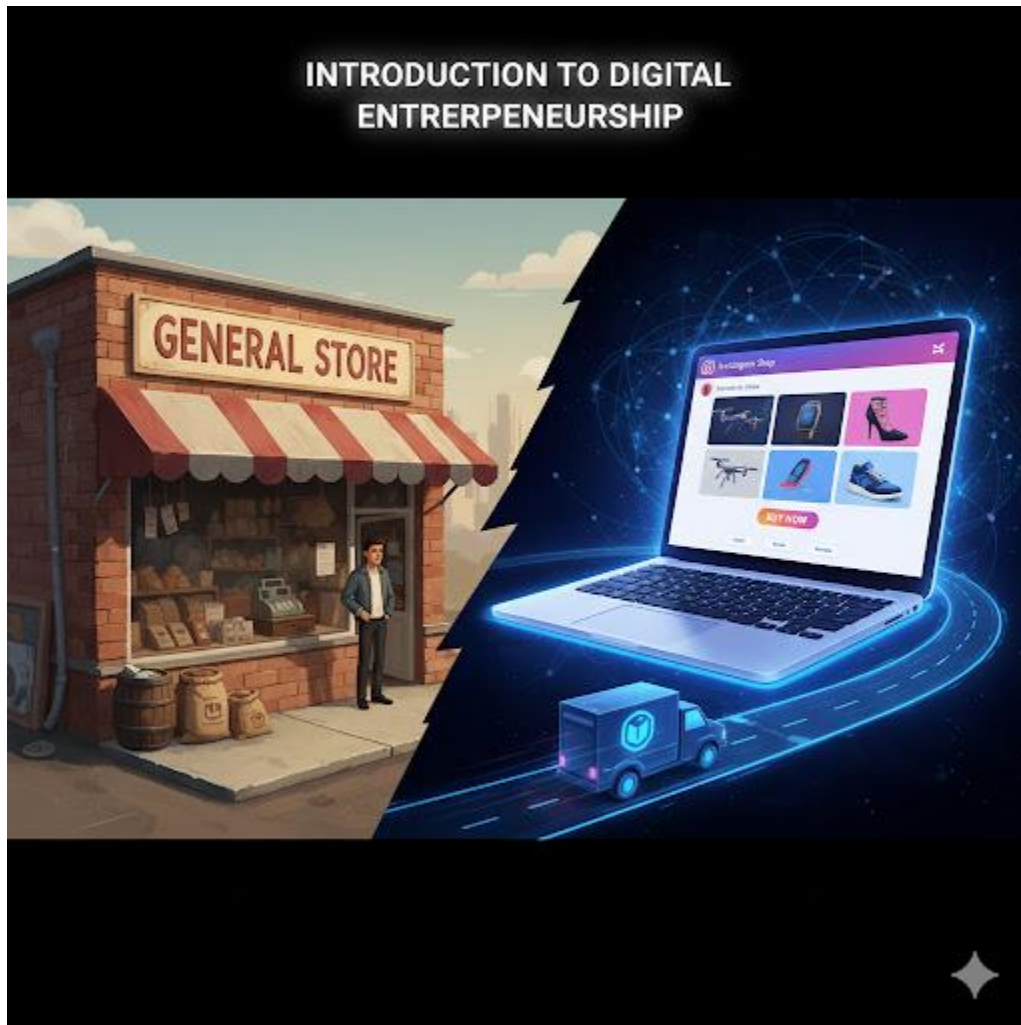
- A quick practical test: "Create a 2-slide PowerPoint. Add a 'Push' transition. On slide 2, make the title 'Fly In'."
- A design challenge: "You have 10 minutes in Canva to create a poster for a school debate."
- A ethics discussion: "Is it 'Fair Use' to use 30 seconds of a Davido song in your personal YouTube video? Why or why not?"



SECOND TERM - JSS 3 DIGITAL TECHNOLOGIES LESSON NOTES



WEEK 1: INTRODUCTION TO DIGITAL ENTREPRENEURSHIP



Content: (I) Meaning and importance (II) Skills and opportunities.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Entrepreneurship" and "Digital Entrepreneurship."
2. Differentiate between a traditional "brick-and-mortar" business and a digital business.
3. State at least three (3) reasons why digital entrepreneurship is important.
4. List three (3) essential skills required for a digital entrepreneur.
5. Identify five (5) digital business opportunities available to a young person.

Lesson Content:

1. What is Entrepreneurship?

- **Definition (Recap):** Entrepreneurship is the process of identifying a business opportunity, taking a financial risk, and organizing resources (capital, labor) to start and run a new business venture with the aim of making a profit.
- An **entrepreneur** is a "go-getter," a problem-solver, and a risk-taker.

2. What is Digital Entrepreneurship?

- **Definition:** Digital Entrepreneurship is the same as regular entrepreneurship, but the business is primarily **online**, using the **internet** and **digital technologies** as its main platform.
- Instead of renting a physical shop, a digital entrepreneur's "shop" is a website, a social media page, or a mobile app.
- **Traditional vs. Digital Business:**
 - **Traditional (Brick-and-Mortar):** A woman who sells shoes in a physical shop at Onitsha Main Market.
 - **Digital:** A woman who sells the same shoes on an Instagram page, a Jumia store, or her own website (e.g., www.shoepalace.ng).

3. Importance of Digital Entrepreneurship

1. **Low Start-up Cost:** It is much cheaper to start an Instagram business page or a simple website than to rent, furnish, and pay for electricity in a physical shop.
2. **Global Market Access:** A physical shop in Lagos can only sell to people who walk past. A digital business on Instagram or a website can sell to anyone in Nigeria, Africa, or the entire world.
3. **Flexibility:** A digital business can be run from anywhere (home, school, a bus) using just a smartphone and an internet connection.
4. **Job Creation:** It creates new types of jobs that did not exist before (e.g., social media manager, content creator, website designer, digital marketer).

5. **Economic Growth:** It allows for faster innovation and contributes billions of dollars to the national economy (e.g., the success of "Fintech" companies like Paystack and Flutterwave).

4. Essential Skills for a Digital Entrepreneur

1. **Digital Literacy (Tech Skills):** You must be comfortable using the tools (e.g., how to post effectively on social media, how to use online payment apps, basic website management).
2. **Digital Marketing:** You must know how to *attract* customers online (this will be covered in Week 4).
3. **Communication:** Writing clear product descriptions, replying to customer emails and DMs (Direct Messages) politely, and creating engaging content.
4. **Financial Management:** Knowing how to price your product, track your sales, and manage your online money.
5. **Resilience and Adaptability:** The digital world changes *very* fast. You must be willing to learn new apps, new trends, and new strategies.

5. Digital Business Opportunities for Students

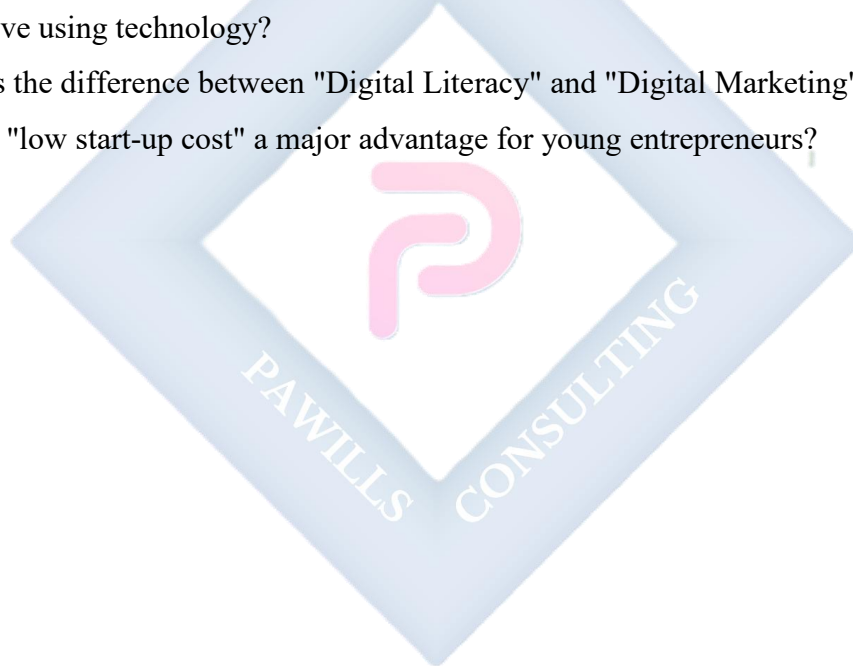
1. **Content Creation:** Starting a blog, a YouTube channel, or a TikTok account about a topic you love (e.g., gaming, comedy, football analysis, science experiments). You make money from adverts.
2. **Social Media Management:** Using your skills to manage the Instagram or Facebook page for a small local business (e.g., your local hairdresser, a restaurant) for a small fee.
3. **E-commerce / Dropshipping:** Selling products (e.g., phone cases, jewelry, t-shirts) on a social media page without needing to keep the stock yourself.
4. **Graphic Design:** Using a tool like Canva (from First Term) to design logos, flyers, and social media posts for people and businesses.
5. **Tutoring:** Creating a small online group (e.g., on WhatsApp or Google Classroom) to teach JSS1 students a subject you are good at, like Mathematics.

Evaluation:

1. What is the main difference between a traditional entrepreneur and a digital entrepreneur?
2. List two (2) reasons why digital entrepreneurship is important.
3. What does it mean to have "Global Market Access"?
4. List three (3) skills a digital entrepreneur needs.
5. List two (2) digital business opportunities a JSS3 student can start.

Assignment:

1. Define Digital Entrepreneurship in your own words.
2. Find a Nigerian business on Instagram (e.g., a food vendor, a fashion store). Write down their name (@). What makes their page look successful?
3. Imagine you want to start a digital business. What "problem" in your community could you solve using technology?
4. What is the difference between "Digital Literacy" and "Digital Marketing"?
5. Why is "low start-up cost" a major advantage for young entrepreneurs?



WEEK 2: INTERNET COMMUNICATION AND COLLABORATION



Content: (I) Email (II) Video calls (III) Cloud tools.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Internet Communication" and "Collaboration."
2. Write and send a professional email, using the "To," "Subject," and "Body" fields correctly.
3. Identify the purpose of video conferencing tools (e.g., Zoom, Google Meet).
4. Define "Cloud Collaboration Tools" and give two examples (e.g., Google Docs, Google Sheets).
5. Explain *how* a tool like Google Docs allows for real-time collaboration.

Lesson Content:

1. What is Internet Communication?

- **Definition:** The exchange of information (text, video, audio) between two or more people over the **internet**.
- It is the backbone of all digital work and social life.

2. What is Collaboration?

- **Definition:** The act of *working together* with one or more people to achieve a common goal (e.g., a school project, a business report).
- **Digital Collaboration:** Using internet-based tools to work together, *even when you are not in the same physical location*.

3. Key Tools for Communication and Collaboration

- **A. Email (Electronic Mail)**
 - **Definition:** A digital messaging system for sending formal text-based messages. It is the "official" communication tool for businesses and schools.
 - **Components of a Professional Email:**
 1. **To:** The recipient's email address (e.g., teacher@school.com).
 2. **Cc (Carbon Copy):** For people who need to *see* the email but don't need to *act* on it (e.g., your HOD).
 3. **Bcc (Blind Carbon Copy):** For people you want to see the email *secretly*. The other recipients will *not* see the Bcc addresses.
 4. **Subject:** A *clear and short summary* of the email's purpose (e.g., "Submission of JSS3 Project" or "Question About Homework"). **Never leave the subject blank.**
 5. **Body:** The main message. It should be polite and clear.
 - Start with a greeting (e.g., "Dear Sir,").
 - Write your message clearly.

- End with a closing (e.g., "Thank you," "Best regards,") and your name.

6. **Attachment:** The button (often a paperclip icon ) used to add a file (like your MS Word document or a photo) to the email.

- **B. Video Conferencing (Video Calls)**

- **Definition:** A tool that allows two or more people in different locations to have a face-to-face meeting in real-time.
- **Purpose:** To have meetings, conduct interviews, or host online classes.
- **Examples:** Zoom, Google Meet, Microsoft Teams, WhatsApp Video Call.
- **Features:** Screen sharing (to show your presentation), chat, recording.

- **C. Cloud Collaboration Tools**

- **Definition:** These are applications that live "in the cloud" (on the internet, not on your computer) and are designed for **real-time teamwork**.
- **Main Example: Google Workspace (formerly G Suite)**
 1. **Google Docs (like MS Word):** A cloud-based word processor.
 2. **Google Sheets (like MS Excel):** A cloud-based spreadsheet.
 3. **Google Slides (like MS PowerPoint):** A cloud-based presentation tool.
- **How They Work (Real-time Collaboration):**
 - You can create a Google Doc for a group project.
 - You "Share" the document with your teammates by sending them a link.
 - **Everyone** can open and *edit the same document at the same time*.
 - You can see your partner's cursor typing *live* on your screen. You can add comments and suggestions.
 - This avoids the old, slow method of: "I do my part in MS Word, save it, email it to you. You add your part, save it as 'Version_2', and email it back."

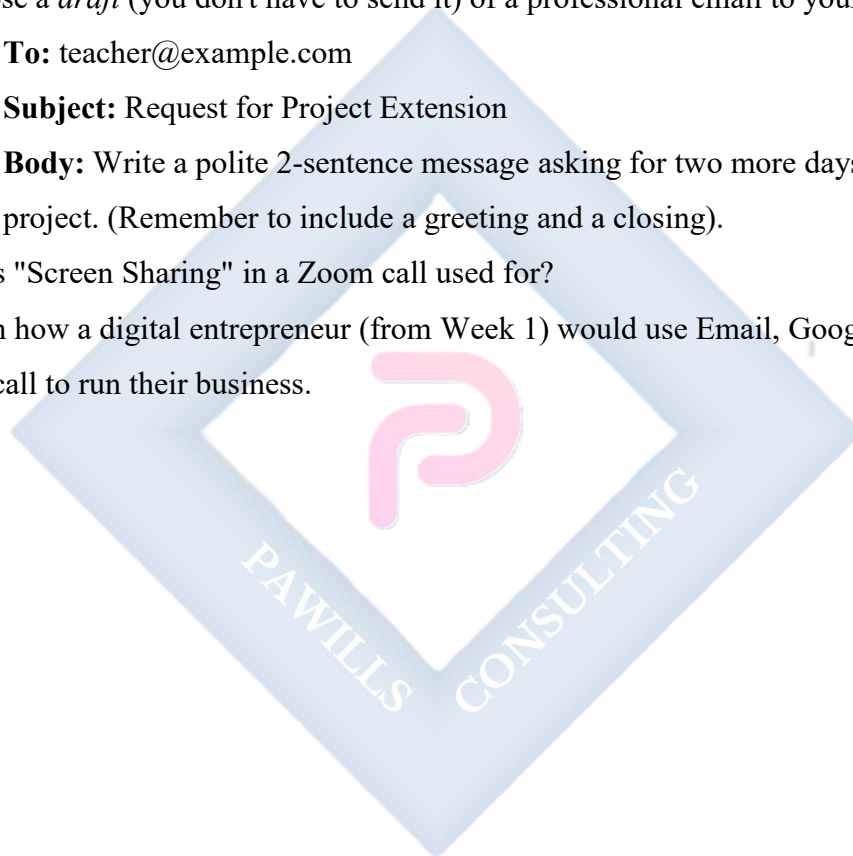
Evaluation:

1. What is "Digital Collaboration"?
2. What is the "Subject" line in an email used for?

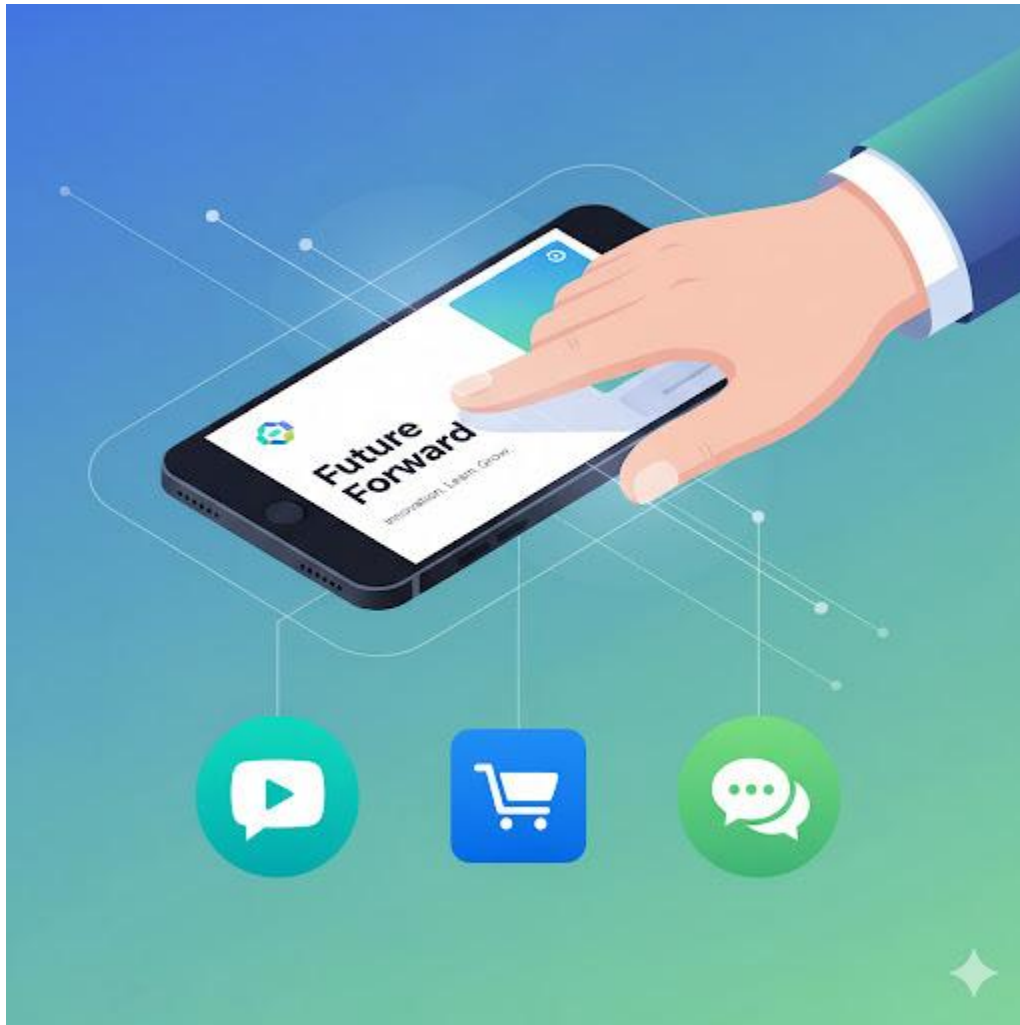
3. What is the difference between "Cc" and "Bcc"?
4. Name two (2) popular video conferencing tools.
5. What is the *main advantage* of using Google Docs for a group project?

Assignment:

1. Define Internet Communication and Digital Collaboration.
2. If you have a GMail account, log in. If not, ask a parent or older sibling.
3. Compose a *draft* (you don't have to send it) of a professional email to your teacher.
 - **To:** teacher@example.com
 - **Subject:** Request for Project Extension
 - **Body:** Write a polite 2-sentence message asking for two more days to finish your project. (Remember to include a greeting and a closing).
4. What is "Screen Sharing" in a Zoom call used for?
5. Explain how a digital entrepreneur (from Week 1) would use Email, Google Sheets, and a Zoom call to run their business.



WEEK 3: SOCIAL MEDIA FOR EDUCATION AND BUSINESS



Content: (I) Responsible use (II) Branding and promotion.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Social Media."
2. Differentiate between the use of social media for "Socializing" vs. "Education" vs. "Business."
3. List three (3) rules for the *responsible* use of social media.
4. Explain how a business can use social media for "Branding."
5. Describe two (2) methods of "Promotion" on social media.

Lesson Content:

1. What is Social Media?

- **Definition:** Websites and applications (apps) that enable users to create and share content (text, photos, videos) and to participate in social networking (connecting with other people).
- **Examples:** Facebook, Instagram, TikTok, X (Twitter), LinkedIn, WhatsApp.

2. Three Uses of Social Media

1. **Socializing (Personal):** The most common use. Connecting with friends and family, sharing personal updates, entertainment (watching videos, memes).
2. **Education (Professional/Academic):** Using platforms to learn.
 - **YouTube:** Watching tutorials on math, science, or how to use Canva.
 - **X (Twitter):** Following experts, organizations (like NASA or WHO), and news outlets for real-time information.
 - **LinkedIn:** A "professional" social media site for building your career and connecting with companies.
3. **Business (Commercial):** Using platforms to find customers, build a brand, and sell products.

3. Responsible Use of Social Media (Digital Citizenship) This is the *ethical* and *safe* way to use social media.

1. **Think Before You Post:** The "Grandma Rule." If you wouldn't want your parent, teacher, or future employer to see it, *do not post it*.
2. **Protect Your Privacy:** Use strong passwords. Don't share sensitive personal data (your home address, phone number, bank details).
3. **Verify Before You Share (Stop Fake News):** Do not spread unverified rumors. You are responsible for the content you share.
4. **No Cyberbullying:** Be respectful. Do not use the platform to harass, insult, or threaten others.

5. **Be Authentic (Avoid "Catfishing"):** Do not use a fake name or someone else's photo to create a fake identity.

4. Social Media for Business: Branding

- **Branding (Recap):** A business's identity.
- Social media (especially Instagram, TikTok, and Facebook) is a powerful branding tool.
- **How to Build a Brand on Social Media:**
 1. **Create a Professional "Bio":** A short, clear description of what your business does and who it serves.
 2. **Visual Consistency:** Use the *same* logo, brand colors, and fonts in all your posts (as learned in First Term). This makes your page look professional and recognizable.
 3. **Tone of Voice:** Decide on your brand's "personality." Are you funny? Formal? Inspiring? Make sure all your captions (the text with your posts) sound consistent.
 4. **Value-Added Content:** Don't just post "Buy my product!" all the time. Give your followers *value*.
 - *Example:* A fashion brand could post "5 Ways to Style a White Shirt."
 - *Example:* A food vendor could post a "1-minute recipe."

5. Social Media for Business: Promotion

- **Promotion:** The act of *advertising* your product or service.
- **Methods of Promotion:**
 1. **Organic Promotion (Free):**
 - Using relevant **hashtags** (e.g., #LagosBaker, #FashionInNigeria) so new people can discover you.
 - Creating "viral" content (e.g., a funny TikTok or an informative Instagram Reel) that people *want* to share.
 - Encouraging "User-Generated Content" (e.g., asking customers to post a photo with your product).
 2. **Paid Promotion (Ads):**

- Paying the platform (e.g., "Boosting" a post on Instagram) to show your ad to thousands of people who match your target audience (e.g., "females, aged 18-25, living in Lagos").

3. Influencer Marketing:

- Paying a popular social media user (an "influencer") to use and showcase your product to their followers.

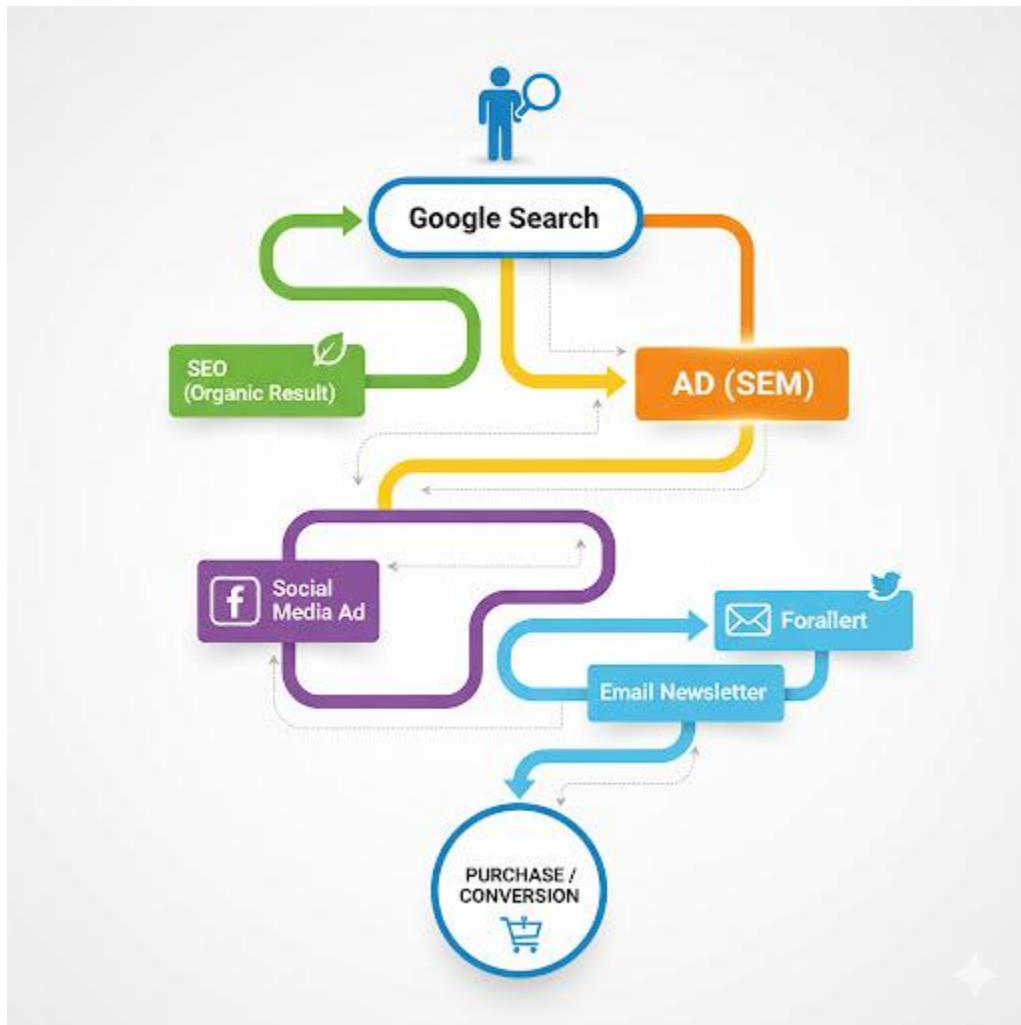
Evaluation:

1. What is social media?
2. What is the "Grandma Rule" for responsible posting?
3. What does it mean to have "visual consistency" in social media branding?
4. What is a "hashtag" used for?
5. What is the difference between "organic promotion" and "paid promotion"?

Assignment:

1. List three (3) social media platforms and their main use (e.g., LinkedIn - professional).
2. What is "catfishing," and why is it an irresponsible use of social media?
3. Explain what "value-added content" is for a business. Give one example for a business that sells skincare products.
4. What is an "influencer"?
5. Search for "How to use social media responsibly." Find one (1) rule or tip that we did not discuss in class.

WEEK 4: DIGITAL MARKETING BASICS



Content: (I) Meaning and channels (II) Advertising safely online.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Digital Marketing."
2. Differentiate between "Digital Marketing" and "Traditional Marketing."
3. List four (4) major channels of digital marketing.
4. Define "SEO" and "Content Marketing."
5. Identify two (2) risks of advertising online and how to stay safe.

Lesson Content:

1. What is Digital Marketing?

- **Traditional Marketing:** Advertising using *offline* channels (e.g., TV commercials, radio jingles, newspaper ads, billboards).
- **Digital Marketing:** The practice of *promoting* products, services, or brands using **digital technologies** and the **internet**.
- It is the "how-to" of finding and communicating with customers online. Social Media Marketing (Week 3) is *one part* of Digital Marketing.

2. Channels of Digital Marketing Digital marketing is not just one thing. It's a system of different channels working together.

1. Social Media Marketing (SMM):

- Using platforms like Instagram, Facebook, TikTok to build a community and run ads.

2. Content Marketing:

- Creating and sharing *valuable, relevant, and consistent* content (like blog posts, videos, podcasts, e-books) to *attract* and *retain* an audience.
- **Goal:** Instead of *pushing* an ad on someone, you *pull* them to you by being helpful and informative. (This is related to "value-added content" from Week 3).

3. Search Engine Optimization (SEO):

- **Definition:** The process of *optimizing* (improving) your website so that it appears at the **top of search engine results** (like Google) for *free* ("organically").
- **How it works:** When you search "best tailor in Abuja" on Google, SEO is the reason one tailor's website is #1 and another is on page 10.
- **Key:** Using the right **keywords** (the words people are searching for) in your website's content.

4. Search Engine Marketing (SEM):

- This is the *paid* version of SEO. It's when you *pay* Google (e.g., Google Ads) to show your website as an "Ad" or "Sponsored" link at the very top of the results page.

5. Email Marketing:

- Building a list of email addresses from interested customers and sending them newsletters, special offers, and product updates.

3. Advertising Safely Online (As a User) The internet is filled with ads. As a citizen, you must be smart and safe.

- **Risk 1: Scams and Fraudulent Ads**

- **The "Too Good to Be True" Rule:** If an ad promises something unbelievable ("Buy one iPhone, get two free!" or "Click here to win \$1,000,000"), it is **100% a scam**.
- **Safety:** Do not click on suspicious links. Do not give your bank details (BVN) or password to any "promo" or "giveaway."

- **Risk 2: Malware and Spyware**

- Some ads (especially "pop-up" ads) contain **malware** (viruses) that can infect your computer if you click them.
- **Safety:** Use an **Ad Blocker** (a browser extension) to block most pop-up ads. Never download "free" software from a site you don't trust.

- **Risk 3: Data Tracking and Privacy**

- Companies (like Facebook and Google) *track* your online behavior (what you search for, what sites you visit) to show you "targeted ads."
- **Safety:** Be aware of this. Read the privacy settings on apps. Use browser "incognito" or "private" modes if you don't want your searches saved.

Evaluation:

1. What is the difference between traditional marketing and digital marketing?
2. What does "SEO" stand for?
3. What is "Content Marketing"?
4. What is the difference between SEO and SEM?
5. What is the "Too Good to Be True" rule for online safety?

Assignment:

1. Define Digital Marketing and list four (4) of its channels.
2. Explain "Content Marketing" in your own words.
3. Go to Google and search for "JSS3 textbooks." Look at the results. Can you identify which links are "Ads" (SEM) and which are "organic" (SEO)?
4. What is an "Ad Blocker," and why is it a useful safety tool?
5. A pop-up ad says your computer has a virus and you must "Click Here to Download Cleaner." What should you do, and why?



WEEK 6: CYBER ETHICS AND NETIQUETTE



Content: (I) Respectful online behavior.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Ethics" and "Cyber Ethics."
2. Define "Netiquette" and explain its importance.
3. List the "10 Core Rules of Netiquette."
4. Analyze a case study of good or bad netiquette.
5. Explain the concept of "flame wars" and how to avoid them.

Lesson Content:

1. What are Ethics?

- **Definition:** Ethics are the *moral principles* that govern a person's behavior or the conducting of an activity. They are the "rules of right and wrong."
- **Example:** It is *unethical* for a teacher to accept a bribe for good grades.

2. What are Cyber Ethics?

- **Definition:** Cyber Ethics are the moral principles and code of conduct for using the internet and digital devices. It is the study of "right and wrong" in the online world.
- **Key Question:** "Just because I *can* do this online, *should* I do it?"

3. What is Netiquette?

- **Definition:** A combination of the words "**Network**" and "**Etiquette**." Netiquette is the set of rules for **polite and respectful behavior** when communicating online.
- **Importance:** In the real world, we see a person's face and hear their tone of voice. Online, we only see *text*. This makes it very easy for misunderstandings to happen. Netiquette provides the "polite" rules to prevent conflict.

4. The Core Rules of Netiquette (Based on the classic rules by Virginia Shea)

1. Rule 1: Remember the Human.

- Do not say anything to someone online that you would not say to their face. There is a real person with feelings behind that screen.

2. Rule 2: Adhere to the same standards of behavior online that you follow in real life.

- Be ethical. Obey the law.

3. Rule 3: Know where you are in cyberspace.

- Netiquette changes from place to place. Behavior that is acceptable in a WhatsApp group with your close friends is *not* acceptable in a formal email to your teacher or in a professional LinkedIn message.

4. Rule 4: Respect other people's time and bandwidth.

- Be brief and to the point. Don't send huge, unnecessary files. Don't "spam" a group chat with 50 unrelated messages.
- 5. **Rule 5: Make yourself look good online.**
 - Write clearly. Use correct spelling and grammar. Know what you're talking about and be logical.
- 6. **Rule 6: Share expert knowledge.**
 - If you know the answer to a question, share it. This is how the internet grows.
- 7. **Rule 7: Help keep "flame wars" under control.**
 - **Flame War:** An angry, insulting, and personal argument that erupts in a public online forum (like a comments section or group chat).
 - **How to stop it:** *Do not participate.* Do not "feed the troll" (reply to the angry person). Ignore them or calmly state the facts.
- 8. **Rule 8: Respect other people's privacy.**
 - Do not "snoop" in someone's files. Do not share someone's private email or photo without their permission.
- 9. **Rule 9: Don't abuse your power.**
 - If you are a system administrator or the "admin" of a group chat, do not abuse your power over other users.
- 10. **Rule 10: Be forgiving of other people's mistakes.**
 - If someone makes a spelling error or a stupid comment, be kind about it. Correct them politely and privately, not by shaming them in public.

5. Other Netiquette Rules

- **DON'T TYPE IN ALL CAPS.** It is interpreted as SHOUTING and is considered rude.
- **Be careful with sarcasm and jokes.** They are often misunderstood in text. Use emojis 😊 to show your tone, but use them professionally.
- **Don't spam.** Do not forward every message, chain letter, or "fake news" you receive.

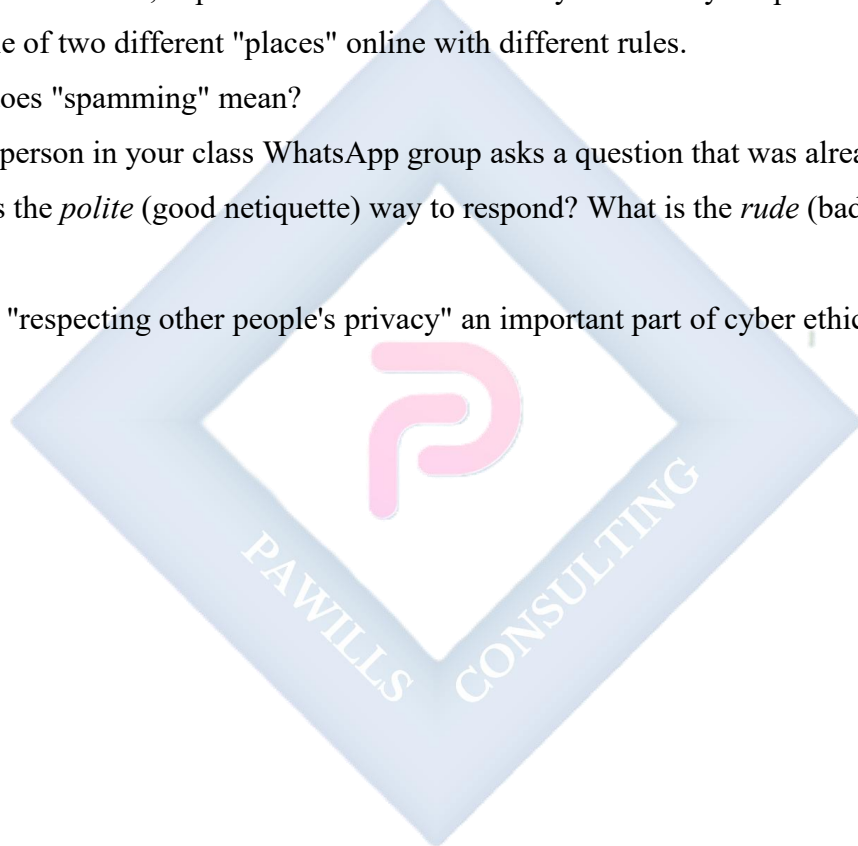
Evaluation:

1. What is the difference between "Ethics" and "Netiquette"?

2. What is the *first* and most important rule of netiquette? (Remember the Human).
3. Why should you avoid typing in ALL CAPS?
4. What is a "flame war"?
5. What is the best way to handle a "flame war"?

Assignment:

1. Define Cyber Ethics and Netiquette.
2. In your own words, explain "Rule 3: Know where you are in cyberspace." Give an example of two different "places" online with different rules.
3. What does "spamming" mean?
4. A new person in your class WhatsApp group asks a question that was already answered. What is the *polite* (good netiquette) way to respond? What is the *rude* (bad netiquette) way?
5. Why is "respecting other people's privacy" an important part of cyber ethics?



WEEK 8: ONLINE PRIVACY AND PROTECTION



Content: (I) Managing personal data (II) Avoiding digital theft.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Online Privacy."
2. Define "Personal Identifiable Information" (PII) and give three examples.
3. Identify three (3) major threats to online privacy (e.g., phishing, malware).
4. Define "Digital Theft" (e.g., identity theft).
5. List five (5) practical steps to protect personal data online.

Lesson Content:

1. What is Online Privacy?

- **Definition:** Online Privacy (or Digital Privacy) is your *right* to control what information about you is collected and shared online, and who has access to it.
- It is the "right to be left alone" in the digital world.

2. What is Personal Identifiable Information (PII)?

- **Definition:** PII is *any* data that can be used to identify *you* specifically. This is the information you must protect.
- **Examples of PII:**
 - Full Name
 - Home Address
 - Phone Number
 - Date of Birth
 - Bank Verification Number (BVN)
 - National Identity Number (NIN)
 - Your Passwords
 - Your mother's maiden name
 - Photos/Videos of you

3. What is Digital Theft?

- **Definition:** Any crime where a thief steals your *digital information* to make money.
- **Identity Theft:** The most common form. A criminal steals your PII (like your name, NIN, and bank details) to *impersonate* you. They can use your identity to take out loans, steal your money, or commit crimes in your name.
- **Phishing:** The *method* used to steal your PII.
 - A scammer sends you a **fake email** or **text message** that *looks* like it's from a real company (e.g., your bank, Netflix, or Google).
 - The message says there is a "problem with your account" and you must "Click this link to log in and fix it."

- The link takes you to a *fake website* (a "phishing site") that looks *identical* to the real one.
- When you type your username and password, you are not giving it to your bank; you are giving it directly to the *thief*.

4. How to Protect Your Privacy and Avoid Theft

1. Use Strong, Unique Passwords.

- **Strong:** At least 12 characters, with uppercase (A), lowercase (a), numbers (1), and symbols (@#\$!).
- **Unique:** Use a *different* password for *every* important account (Email, Bank, Social Media).

2. Enable Two-Factor Authentication (2FA).

- This is the *best* protection. Even if a thief steals your password, they *also* need a second code (sent to your phone) to log in.

3. Think Before You Click (Avoid Phishing).

- *Always* be suspicious of urgent emails.
- *Never* click a link in an email to log in to your bank. Instead, type the bank's real website address (www.gtbank.com) into your browser *manually*.

4. Manage Your Privacy Settings.

- Go into the settings of Instagram, Facebook, and WhatsApp. Set your profile to "Private."
- Control who can see your posts and tag you.

5. Be Wary of Public Wi-Fi.

- "Free" Wi-Fi (at airports, hotels, cafes) is *not* secure.
- Avoid logging into your bank or shopping online when using public Wi-Fi. A hacker on the same network can steal your data.

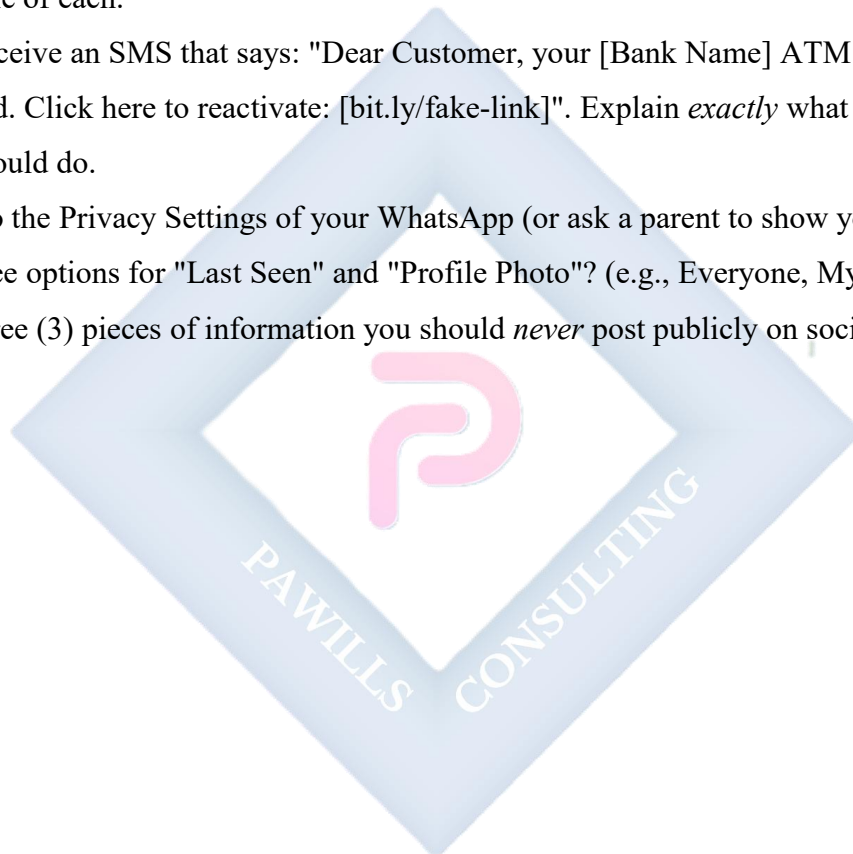
Evaluation:

1. What is "Online Privacy"?
2. What does "PII" stand for? Give two examples.

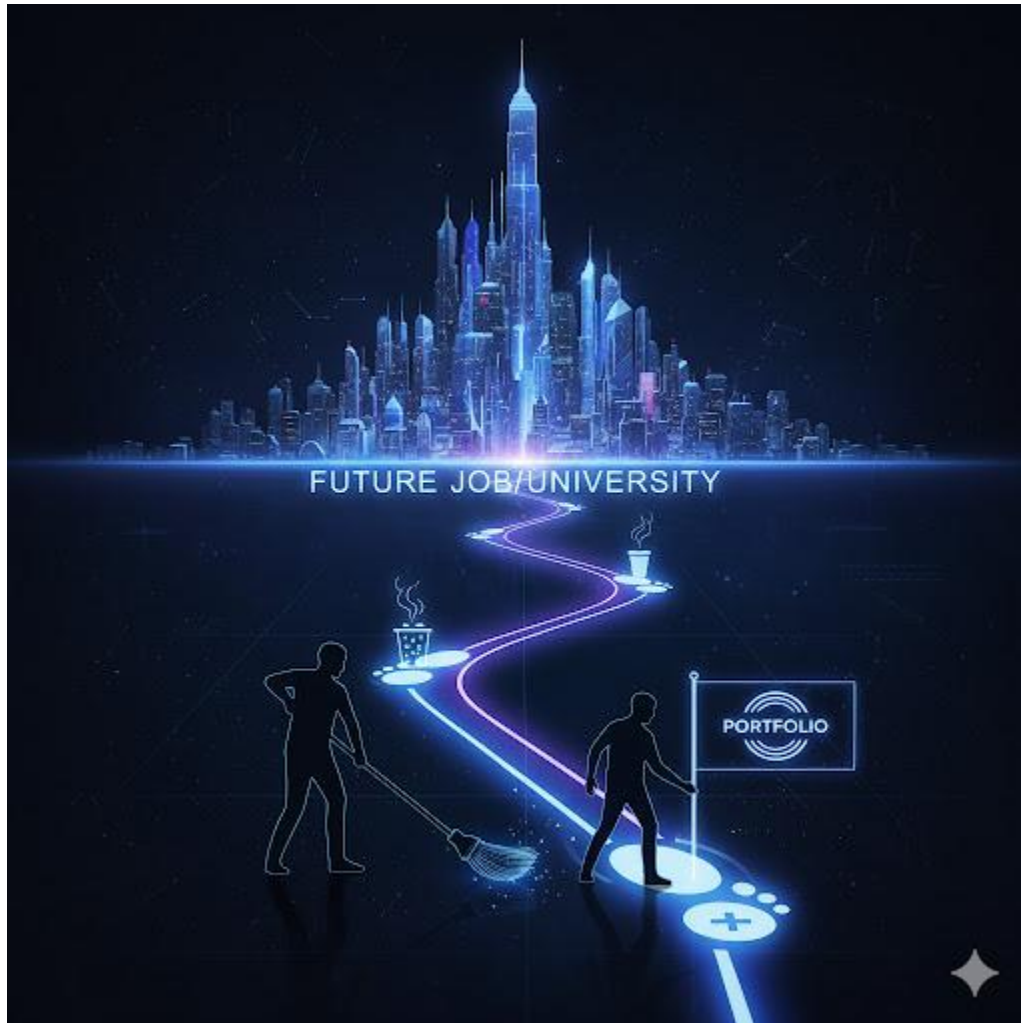
3. What is "Phishing"?
4. What is "Two-Factor Authentication (2FA)"?
5. Why is it a bad idea to use the same password for your email and your bank?

Assignment:

1. Define Online Privacy and Digital Theft.
2. What is the difference between a "strong" password and a "weak" password? Give an example of each.
3. You receive an SMS that says: "Dear Customer, your [Bank Name] ATM card has been blocked. Click here to reactivate: [bit.ly/fake-link]". Explain *exactly* what this is and what you should do.
4. Go into the Privacy Settings of your WhatsApp (or ask a parent to show you). What are the three options for "Last Seen" and "Profile Photo"? (e.g., Everyone, My Contacts, ...).
5. List three (3) pieces of information you should *never* post publicly on social media.



WEEK 9: MANAGING DIGITAL FOOTPRINT



Content: (I) Meaning and importance (II) Creating a positive online presence.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Digital Footprint."
2. Differentiate between a "Passive" and an "Active" digital footprint.
3. Explain *why* their digital footprint is important (e.g., for future jobs, university).
4. List three (3) ways to "clean up" a digital footprint.
5. List three (3) ways to create a *positive* digital footprint.

Lesson Content:

1. What is a Digital Footprint?

- **Definition:** Your Digital Footprint is the **trail of data** you leave behind every time you use the internet. It is the "footprint" that shows where you have been and what you have done online.
- It includes:
 - Websites you visit
 - Social media posts, likes, and comments
 - Photos and videos you upload
 - Your search history
 - Online purchases
 - Emails you send
- **Key Fact:** Your digital footprint is **permanent**. Even if you "delete" a post, it may still exist on a server or in someone's screenshot.

2. Types of Digital Footprints

- **A. Active Digital Footprint:**
 - This is the data you create *intentionally* and *knowingly*.
 - **Examples:**
 - Sending an email.
 - Uploading a photo to Instagram.
 - Tweeting a comment.
 - Filling out an online form.
- **B. Passive Digital Footprint:**
 - This is the data trail you leave *unintentionally* and *without awareness*.
 - **Examples:**
 - Websites using "cookies" to track what you look at.
 - Apps (like Instagram) collecting data on your location.
 - Your "search history" in Google.

3. Why is Your Digital Footprint Important?

- **Your Online Reputation:** Your footprint is your "digital reputation."
- **Future Opportunities:**
 - **University Admissions:** Admissions officers *will* Google you. They *will* look at your social media. A profile full of cyberbullying, cheating, or inappropriate photos can get your admission *denied*.
 - **Future Employers:** When you apply for a job, the *first thing* the employer will do is search your name online. Your digital footprint is your new resume.
- **Personal Security:** A large, messy footprint can make you an easy target for identity theft.

4. How to Manage and "Clean Up" Your Footprint

1. **Google Yourself:** Search your own full name (in "incognito" mode) and see what comes up. This is what the world sees.
2. **Delete Old/Embarrassing Posts:** Go back through your old social media (Instagram, Facebook, X). Delete any posts, photos, or comments that are rude, silly, or unprofessional.
3. **Check Your Privacy Settings:** (As in Week 8). Make all your *personal* social media accounts "Private" (for Friends Only).
4. **Unsubscribe:** Unsubscribe from email newsletters you no longer read.
5. **Use a "Professional" Email:** Your email for official applications should be simple (e.g., Bisi.Adekunle@gmail.com, not LagosBabe123@yahoo.com).

5. How to Create a *Positive* Digital Footprint This is the *best* strategy. Instead of just hiding the bad, you should *create good*.

1. **Use LinkedIn:** Create a professional profile on LinkedIn (when you are older) to showcase your skills and school projects.
2. **Create a Digital Portfolio (from First Term):** This is the *best* way to control your online reputation. When someone Googles you, your professional portfolio (on Google Sites) will be the first thing they see.

3. **Share Your Achievements:** Post about academic competitions you won, useful projects you built, or volunteer work you have done.
4. **Be a "Creator," Not Just a "Commenter":** Write a positive blog post, create an informative video, or share helpful resources.

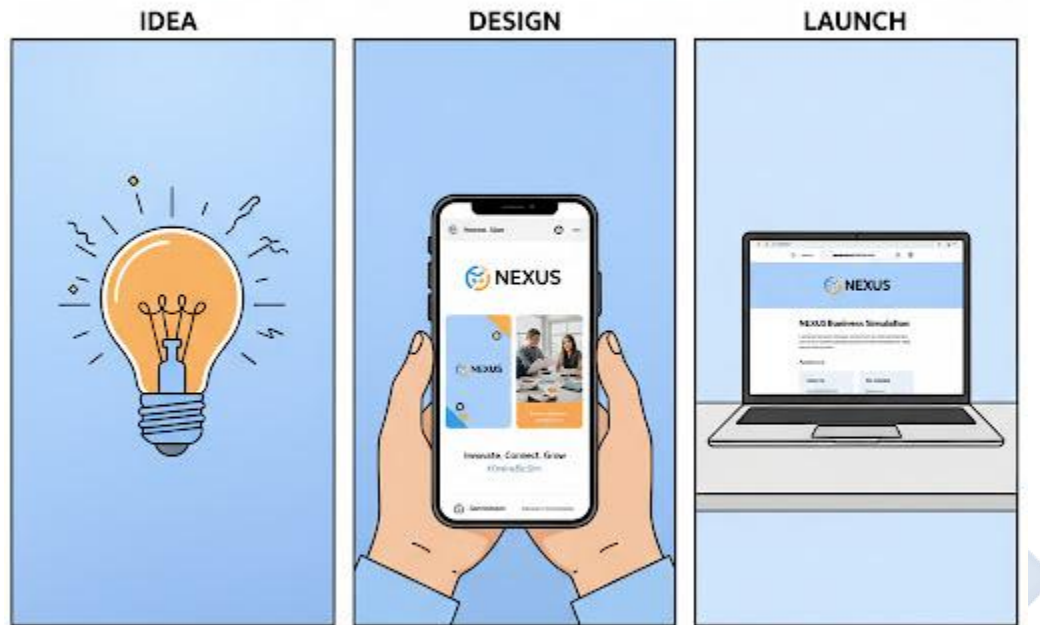
Evaluation:

1. What is a "digital footprint"?
2. What is the difference between an "active" and a "passive" digital footprint?
3. Give one example of an *active* footprint.
4. Give one example of a *passive* footprint.
5. Why is it important to have a *positive* digital footprint?

Assignment:

1. Define Digital Footprint in your own words.
2. Why is your digital footprint considered "permanent"?
3. Go to Google and search your own full name (with your parents' permission). What is the first result that comes up?
4. List two (2) things you can do *today* to "clean up" or "manage" your digital footprint.
5. List two (2) things you can do to build a *positive* digital footprint.

WEEK 10: DIGITAL PROJECT: ONLINE BUSINESS SIMULATION



Content: (I) Students create mock e-business or blog.

Performance Objectives: By the end of this lesson, students should be able to:

1. Integrate skills from the entire term (branding, marketing, collaboration).
2. Develop a simple business idea.
3. Create a 1-page "landing page" or "mock blog" for their business.
4. Create one (1) sample social media post (using Canva) to promote their business.
5. Present their mock business to the class.

Lesson Content:

1. Project Objective To combine all the skills learned this term into one practical project. You will *simulate* (pretend to create) a digital business.

- **Skills to use:**
 - **Week 1 (Digital Entrepreneurship):** Your business idea.
 - **Week 2 (Collaboration):** Work in small groups (optional).
 - **Week 3 (Social Media):** Create a sample promo post.
 - **First Term (Canva/Branding):** Design a logo and promo post.
 - **First Term (Digital Portfolio):** Use a tool like Google Sites to build a "landing page" (a simple 1-page website).

2. Project Steps (In-Class Activity)

- **Step 1: Brainstorm Your Business (10 mins)**
 - Get into groups of 2-3.
 - Choose a simple digital business idea (from Week 1).
 - *Examples:* "Aisha's Cakes" (sells cakes via Instagram), "JSS3 Tutors" (a blog that shares study tips), "Naija Kicks" (sells custom-painted sneakers), "Tech-Fix" (a blog that reviews phone apps).
 - **Decide:**
 1. Business Name
 2. What do you sell/offer?
 3. Who is your target customer? (e.g., students, parents, fashion lovers).
- **Step 2: Create Your Brand (20 mins)**
 - Go to **Canva.com**.
 - Design a simple **Logo** for your business. (Search "Logo" templates).
 - Choose your 2 **Brand Colors**.
 - Design one (1) sample **Social Media Post** (use "Instagram Post" template) that advertises your product.
 - Download your Logo and your Post as PNG/JPG files.
- **Step 3: Build Your "Shop" (Landing Page) (40 mins)**
 - Go to **sites.google.com**.

- Create a "New Site."
- **On this one page, you must add:**
 1. **A Title:** Your Business Name.
 2. **Your Logo:** (Use Insert > Image).
 3. **A Slogan/Intro:** (Use Insert > Text Box). Write one sentence describing what you do. (e.g., "The best homemade cakes in Lagos!" or "Your #1 source for JSS3 study tips.").
 4. **A Sample Product/Post:** (Use Insert > Content Block). Upload your Canva promo post on one side, and write a description on the other.
 5. **"How to Order" / "Contact Us":** (Use Insert > Text Box). Write "To order, send a DM to our Instagram @[YourFakeBizName] or email [YourFakeEmail]."
- Click **Publish** and give it a site name.
- **Step 4: Presentation (Next Class)**
 - Each group presents their Google Site to the class (1-2 minutes).
 - They must explain their business idea, their brand, and their marketing post.

Evaluation (Project-Based):

1. (Presentation) Did the group have a clear business idea?
2. (Canva) Is the logo and promo post clear and consistent?
3. (Google Sites) Did the landing page contain all the required elements (Logo, Title, Text, Post)?
4. (Marketing) Is the promo post engaging?
5. (Collaboration) Did the group work well together?

Assignment (Continuation of Project):

1. Complete your Google Site for the project.
2. Copy the *published* link (URL) of your Google Site.
3. Compose a new, professional **Email** (as in Week 2) to your teacher.
 - **Subject:** Digital Project Submission - [Your Business Name]

- **Body:** "Dear [Teacher's Name], Please find the link to our online business simulation project: [Paste your Google Site link here]. Best regards, [Your Names]."
4. (If you have time) Create a *second* promo post in Canva.
 5. What was the *hardest* part of this project? (e.g., deciding the idea, designing the logo, building the site).



WEEK 11: REVISION



Content: Review of term topics.

Note: This week is dedicated to a comprehensive review of all topics covered in the Second Term, especially in preparation for the Basic Education Certificate Examination (BECE).

Revision Checklist:

1. **Digital Entrepreneurship (Week 1):**
 - Traditional vs. Digital business.

- Importance: Low cost, global market, flexibility.
 - Skills: Digital literacy, marketing, communication.
 - Opportunities: Content creation, e-commerce, social media management.
2. **Internet Communication (Week 2):**
- **Email:** Professional use (To, Subject, Body, Attachment).
 - **Video Calls:** Zoom, Google Meet (Screen sharing).
 - **Cloud Collaboration:** Google Docs (real-time editing).
3. **Social Media (Week 3):**
- Uses: Social, Education, Business.
 - **Responsible Use:** Think before you post, no cyberbullying, no fake news.
 - **Branding:** Visual consistency, value-added content.
 - **Promotion:** Organic (hashtags) vs. Paid (ads) vs. Influencers.
4. **Digital Marketing (Week 4):**
- Channels: SMM, Content Marketing, Email Marketing.
 - **SEO (Search Engine Optimization):** Ranking high on Google for *free*.
 - **SEM (Search Engine Marketing):** *Paying* Google for ads.
 - **Safety:** The "Too Good to Be True" rule, avoiding scams, malware.
5. **Cyber Ethics & Netiquette (Week 6):**
- **Ethics:** Moral rules ("Should I do this?").
 - **Netiquette:** Polite behavior ("Remember the Human").
 - **Rules:** No ALL CAPS, no spam, no flame wars, respect privacy.
6. **Online Privacy (Week 8):**
- **PII (Personal Identifiable Information):** Data that identifies you (NIN, BVN, address).
 - **Digital Theft:** Identity theft.
 - **Phishing:** Fake emails that steal your password.
 - **Protection:** Strong/unique passwords, 2FA, check settings.
7. **Digital Footprint (Week 9):**
- **Definition:** Your "digital trail" of data.
 - **Active (Intentional):** Posting, emailing.
 - **Passive (Unintentional):** Cookies, search history.

- **Importance:** Permanent; affects future university/job applications.
- **Positive Footprint:** Create a portfolio (Google Sites), post achievements.



THIRD TERM - JSS 3 DIGITAL TECHNOLOGIES LESSON NOTES



WEEK 1: MANAGEMENT OF E-WASTE



Content: (I) Meaning and sources of e-waste (II) Safe disposal methods (III) Environmental effects.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "e-waste" (Electronic Waste).
2. List at least five (5) common sources of e-waste.
3. Explain the serious environmental effects of improper e-waste disposal.
4. Describe at least three (3) safe methods for managing e-waste.
5. Differentiate e-waste from general (biodegradable) waste.

Lesson Content:

1. What is e-Waste?

- **Definition: e-Waste** (Electronic Waste) is a term for all **discarded, obsolete, or broken electrical or electronic devices**.
- It is any household or office appliance with a battery or plug that has reached the end of its "useful life."
- e-Waste is the fastest-growing type of waste in the world due to the high demand for new technology.

2. Common Sources of e-Waste e-Waste is generated from a wide range of products:

1. **Communication Devices:** Smartphones (e.g., old Tecno, iPhone), feature phones, landlines, modems, routers.
2. **Computing Devices:** Laptops, desktop computers (CPUs, monitors), tablets, printers, scanners, keyboards, mice, hard drives.
3. **Entertainment Electronics:** Televisions (especially old CRT "box" TVs), DVD players, CD players, game consoles (e.g., PlayStation, Xbox), radios.
4. **Household Appliances:** Microwaves, blenders, refrigerators, air conditioners, electric fans, washing machines.
5. **Power & Accessories:** Batteries (phone, laptop, inverter), power banks, chargers, USB cables, stabilizers (UPS).
6. **Lighting:** Fluorescent tubes, LED bulbs.

3. Environmental and Health Effects of Improper Disposal e-Waste is **extremely hazardous** because it contains toxic heavy metals and chemicals. When dumped in regular landfills or burned, these toxins poison the environment and people.

- **Location:** Places like Alaba Market or Computer Village in Lagos, which have large informal repair/disposal markets, are at high risk.
- **Major Toxins and Their Effects:**

1. **Lead (Pb):** Found in old TV/computer monitors (CRT) and solder. It is a potent neurotoxin that damages the brain, nervous system, and kidneys, especially in children.
 2. **Mercury (Hg):** Found in fluorescent bulbs and flat-screen (LCD) monitors. It damages the brain and kidneys. When burned, it enters the air and falls into rivers, poisoning fish and the people who eat them.
 3. **Cadmium (Cd):** Found in old rechargeable batteries and semiconductors. It is a carcinogen (causes cancer) and severely damages the lungs and kidneys.
 4. **Brominated Flame Retardants (BFRs):** Chemicals added to plastic casings to prevent fire. When burned, they release dioxins and furans, which are highly toxic, cause cancer, and disrupt human hormones.
- **Environmental Impact:**
 - **Water Pollution:** When rain falls on an e-waste dump, these heavy metals **leach** (dissolve and seep) into the soil, contaminating groundwater and nearby rivers. This kills aquatic life and makes water unsafe for drinking.
 - **Air Pollution:** The common practice of burning e-waste (e.g., burning wires to get the copper inside) releases a black, toxic smoke. This smoke contains carcinogens and heavy metals, causing severe respiratory problems (like asthma) for people nearby.

4. Safe Disposal and Management Methods e-Waste must *never* be thrown in the regular bin or burned.

1. **The 3 Rs (Reduce, Reuse, Recycle):**
 - **Reduce:** The *best* method. Use your devices for as long as possible. Don't upgrade your phone every year if it still works.
 - **Reuse / Refurbish:** (See Week 2). Repair broken devices. Donate your old, working laptop to a school or a student who needs it.
 - **Recycle (Formal):** This is the most important disposal method.
2. **Formal e-Waste Recycling:**

- This involves sending the e-waste to a specialized factory (not an open-air market).
- In the factory, the devices are safely dismantled.
- Toxic parts (like batteries) are separated and treated.
- Valuable materials (gold, copper, silver, aluminum, plastic) are recovered and can be used to make new products.

3. **Take-Back Programs:**

- Many large manufacturers (like HP, Dell, Apple) and telecommunication companies (like MTN) have programs where you can return your old, "end-of-life" product to them. They are then responsible for recycling it safely.

4. **Government Regulation:**

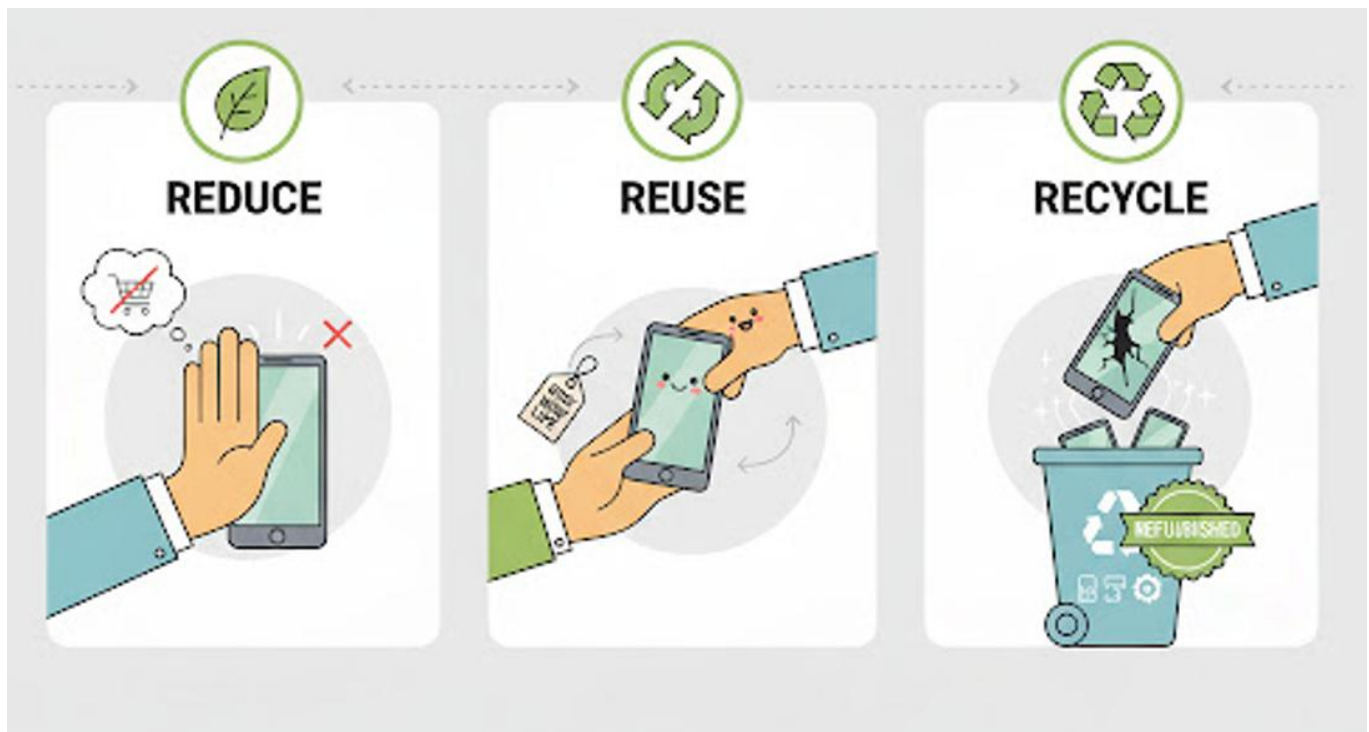
- Governments (like through NESREA in Nigeria) must create laws that *ban* the dumping of e-waste and *support* the creation of formal, safe recycling companies.

Evaluation:

1. What is "e-waste"?
2. List three (3) sources of e-waste in your home.
3. Name two (2) toxic heavy metals found in e-waste.
4. Explain what "leaching" is and how it pollutes water.
5. What is the *best* and safest method for managing e-waste? (Formal Recycling).

Assignment:

1. Define e-Waste in your own words.
2. What is the difference between e-waste and regular waste (like a banana peel)?
3. Explain the environmental danger of *burning* e-waste.
4. What are "Take-Back Programs"?
5. Conduct a small e-waste audit at home (with your parents' permission). Count how many old, broken, or unused electronic devices (phones, chargers, remotes) are in your house. What is the current plan for disposing of them?



WEEK 2: RECYCLING DIGITAL DEVICES

Content: (I) Refurbishing and reusing old gadgets.

Performance Objectives: By the end of this lesson, students should be able to:

1. Re-state the "3 Rs" of waste management.
2. Define "Reusing" in the context of digital devices.
3. Define "Refurbishing" and differentiate it from "Repairing."
4. Explain the economic and environmental benefits of refurbishing.
5. Identify the dangers of informal "recycling" practices.

Lesson Content:

1. The 3 Rs of e-Waste Management (This is the guiding principle for managing all waste, especially e-waste)

- **1. REDUCE:** The best solution. Buy fewer gadgets. Use your phone for 4 years, not 1.
- **2. REUSE:** The second-best solution. If a gadget still works, let someone else use it.
- **3. RECYCLE:** The last and final solution. If a gadget is truly broken and cannot be reused, break it down to recover its raw materials.

2. Understanding "Reuse"

- **Definition:** "Reuse" is the act of passing a product on to a second, third, or fourth user with *minimal or no changes*.
- **It extends the life of a product.**
- **Examples:**
 - Giving your old (but working) smartphone to a younger sibling or a parent.
 - Donating old (but working) computers from an office to a school.
 - Selling your old phone on Jiji or to a repair shop.
- **Benefit:** This is highly sustainable. It keeps a functional product out of the landfill and provides a low-cost device for someone who may not afford a new one.

3. Understanding "Refurbishing"

- **Definition:** "Refurbishing" is a formal process of *rebuilding* a used, returned, or slightly faulty product to a "like-new" condition.
- **Refurbishing vs. Repairing:**
 - **Repairing:** Fixes one specific problem (e.g., "My screen is cracked, replace it").
 - **Refurbishing:** A *complete* overhaul. The device is:
 1. **Tested:** A full diagnostic is run.
 2. **Cleaned:** All dust is removed, and the casing is cleaned.
 3. **Repaired:** Any faulty parts (like an old battery or a slow hard drive) are *replaced*.
 4. **Wiped:** All personal data from the previous owner is securely erased (a "factory reset").
 5. **Certified:** The device is tested again and often sold with a new, short warranty (e.g., "6-Month Certified Refurbished").

- **Examples:** "Refurbished iPhones," "Refurbished Laptops." These are very common in markets like Computer Village.

4. Benefits of Reusing and Refurbishing

- **Environmental Benefits:**
 - **Reduces e-Waste:** Every refurbished laptop sold is one *less* laptop in a landfill.
 - **Saves Energy:** It takes far more energy (electricity, oil, gas) to *manufacture a new laptop* from raw materials than to refurbish an old one.
 - **Conserves Resources:** It reduces the need to mine new gold, copper, and coltan.
- **Economic Benefits:**
 - **Affordability:** Provides high-quality, reliable devices to low-income students, families, and businesses.
 - **Job Creation:** Creates a large and skilled industry of technicians, repair specialists, and salespersons.

5. The Dangers of *Informal* Recycling

- **The Problem:** True, formal recycling (as discussed in Week 1) is expensive and requires a factory. In many parts of Nigeria, "recycling" is done *informally*.
- **Informal Recycling:** This involves untrained people (often children) in open-air scrap yards (like Agbogbloshie in Ghana or parts of Alaba in Lagos) breaking e-waste apart with hammers and burning it to recover valuable metals.
- **The Dangers:**
 - **Extreme Health Hazard:** As seen in Week 1, these workers are exposed *directly* to lead, mercury, and cadmium. They breathe toxic smoke daily, leading to severe lung damage, brain damage, and cancer.
 - **Environmental Devastation:** The ground and water in these areas become "dead zones," so poisoned with heavy metals that nothing can grow, and the water is undrinkable.
- **Solution:** We must move from this dangerous *informal* practice to *formal* refurbishing and recycling industries.

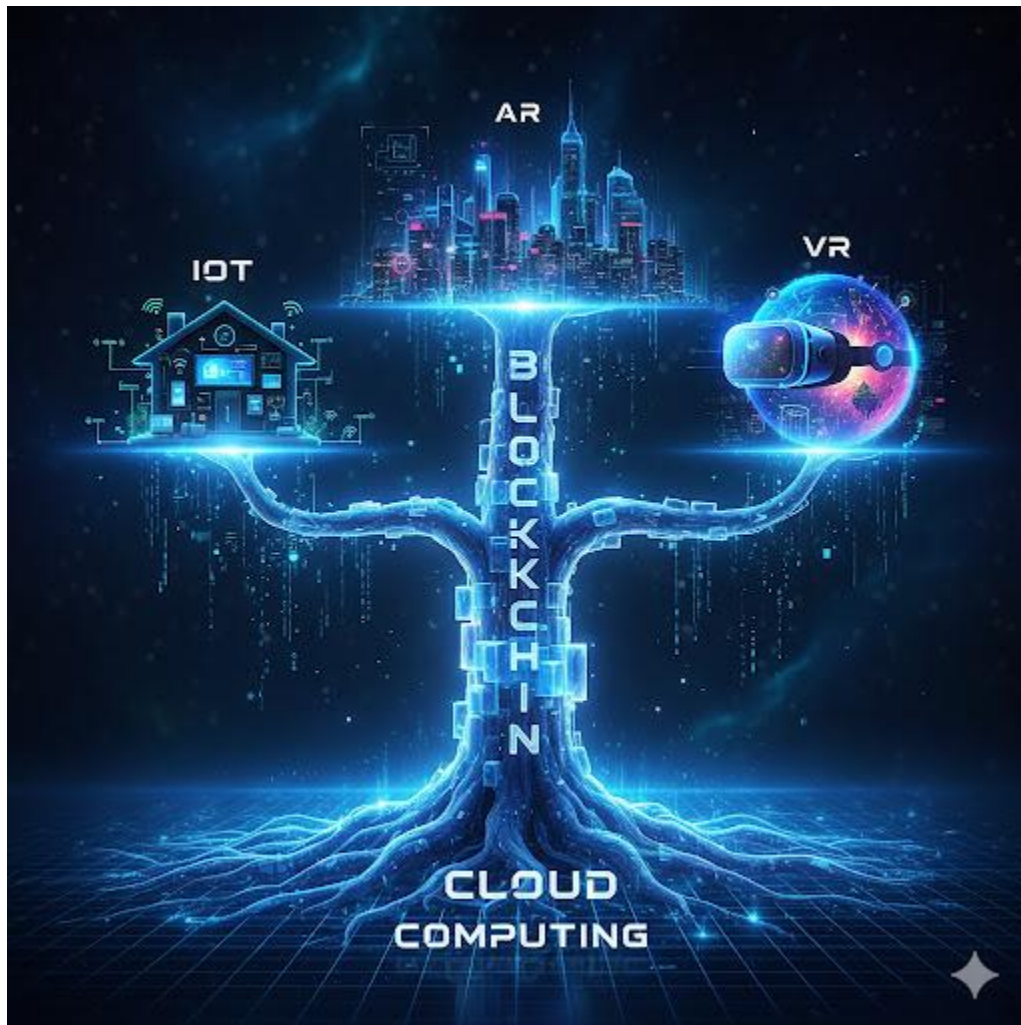
Evaluation:

1. What are the "3 Rs" of waste management?
2. What is the difference between "Reusing" and "Refurbishing"?
3. What is the difference between "Refurbishing" and "Repairing"?
4. List two (2) environmental benefits of refurbishing old computers.
5. What is "informal recycling," and why is it extremely dangerous?

Assignment:

1. Define Reuse, Refurbish, and Recycle.
2. You have a laptop with a slow hard drive and a weak battery, but it works. Would you "repair" it or "refurbish" it? Explain your choice.
3. A large office is upgrading to new computers. They have 100 old, working computers. What is the *most sustainable* (best for the environment and society) thing they could do with them?
4. Why is "Reduce" the most important of the 3 Rs?
5. Search online (with parents' permission) for "Agbogbloshie e-waste." Write two sentences about what you see or read.

WEEK 3: EMERGING TECHNOLOGIES



Content: (I) IoT, Cloud, Blockchain, AR/VR.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Emerging Technologies."
2. Define the "Internet of Things" (IoT) and give one example.
3. Define "Cloud Computing" and name one service (e.g., Google Drive).
4. Define "Blockchain" at a basic level (a secure, shared ledger).
5. Differentiate between "Augmented Reality" (AR) and "Virtual Reality" (VR).

Lesson Content:

1. What are Emerging Technologies?

- **Definition:** "Emerging Technologies" are new technologies whose development and application are still in their early stages, but they are expected to have a **significant impact** on society, business, and our daily lives in the near future.
- These are the "next big things" in the digital world.

2. Key Emerging Technologies

• A. The Internet of Things (IoT)

- **Definition:** A giant network of *physical objects* (devices, vehicles, home appliances) that are embedded with **sensors** and **software** that allow them to **connect to the internet** to send and receive data.
- **Simple Meaning:** Giving everyday "dumb" objects the power of the internet, making them "smart."
- **Examples:**
 - **Smartwatch:** An IoT device. It uses sensors to track your heart rate and sends the data to your phone via the internet.
 - **Smart Home:** A "smart bulb" you can turn on/off with your phone from anywhere, or a "smart fridge" that tells you when you are out of milk.
 - **Smart City:** Using IoT sensors to manage traffic lights, monitor pollution, or detect water leaks.

• B. Cloud Computing

- **Definition:** (Recap from JSS3 Term 1) The delivery of "on-demand" computing services—including servers, storage, databases, and software—over the **Internet** (the "cloud").
- **Simple Meaning:** Using someone else's powerful computer (a "server" in a data center) through the internet, instead of using your own.
- **You use it every day:**
 - **Cloud Storage:** Google Drive, iCloud, Dropbox.
 - **Cloud Apps:** Canva, Google Docs, GMail, Netflix. (The software runs on their server, not your device).

- **Importance:** It allows small businesses (digital entrepreneurs) to access powerful tools (like databases and servers) without having to buy expensive hardware.
- **C. Blockchain**
 - **Definition:** A decentralized, distributed, and immutable digital ledger.
 - **Breakdown:**
 - **Ledger:** A book for recording transactions (like a bank's record book).
 - **Digital:** It's on a computer.
 - **Distributed:** It is not owned by one person or bank. *Everyone* on the network has an *identical copy* of the ledger.
 - **Decentralized:** There is no central boss (like a bank manager). The network is run by all its members.
 - **Immutable:** "Immutable" means **unchangeable**. Once a transaction ("block") is added to the "chain," it *cannot* be altered or deleted, ever. This makes it highly secure and transparent.
 - **Use:** Its most famous use is for **Cryptocurrency** (like Bitcoin), but it can also be used for secure voting, tracking land ownership, or verifying the source of a product (e.g., "Is this drug fake?").
- **D. Augmented Reality (AR) vs. Virtual Reality (VR)**
 - **Augmented Reality (AR):**
 - **Definition:** *Overlays* digital information (images, text) onto the *real world* through your device (usually a phone camera). It *adds to* (augments) reality.
 - **Example:**
 - **Snapchat / Instagram Filters:** They "augment" your real face with digital dog ears.
 - **Pokémon Go:** You see the real street through your camera, but a digital Pokémon is *overlaid* on top.
 - **Google Maps Live View:** You hold up your phone, and it overlays a digital arrow onto the real street to show you where to turn.
 - **Virtual Reality (VR):**

- **Definition:** *Replaces* your reality with a completely **immersive, 3D, computer-generated environment**. You are "transported" to a new world.
- **Requires:** A **VR headset** (like the Meta Quest or PlayStation VR) that blocks your view of the real world.
- **Example:** A VR game where you are *inside* the game, able to look around 360 degrees. A VR simulation where a medical student can "practice" surgery.

Evaluation:

1. What is an "Emerging Technology"?
2. What does "IoT" stand for? Give one example.
3. What does it mean for a "Blockchain" to be "immutable"?
4. What is the *main* difference between AR and VR?
5. Is a Snapchat filter an example of AR or VR?

Assignment:

1. Define IoT, Blockchain, AR, and VR.
2. What is the difference between "Cloud Computing" and "Cloud Storage"?
3. Name two (2) advantages of using IoT in a "Smart Home."
4. Why is Blockchain considered highly secure?
5. Which technology do you think will have the *biggest* impact on Nigeria in the next 10 years, and why? (IoT, AI, Blockchain, or VR?)

WEEK 4: PREPARING FOR DIGITAL FUTURE



Content: (I) STEM education (II) Career pathways.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define the acronym "STEM."
2. Explain why STEM education is critical for national development.
3. List at least five (5) career pathways in the digital technology sector.
4. Describe the roles of a "Software Developer" and a "Cybersecurity Analyst."
5. Identify the non-technical ("soft") skills required for a digital career.

Lesson Content:

1. The Digital Future

- The "future" is digital. Almost *every* job, from banking and medicine to farming and fashion, is being transformed by technology (AI, IoT, Cloud Computing).
- To be successful in this future, a new set of skills is required.

2. What is STEM Education?

- **Definition:** STEM is an acronym for a broad field of study:

- **S** - Science (e.g., Biology, Chemistry, Physics)
- **T** - Technology (e.g., Programming, Network Admin, Digital Design)
- **E** - Engineering (e.g., Civil, Electrical, Robotics, Chemical)
- **M** - Mathematics

- **Why is STEM Important?**

1. **It Teaches Problem-Solving:** STEM is **not** about memorizing facts; it's about learning *how to think critically and solve complex problems*.
2. **It Drives Innovation:** All emerging technologies (AI, IoT, Robotics) are *created* by STEM professionals.
3. **National Development:** A country cannot develop without its own doctors, engineers, programmers, and scientists. Nigeria *must* produce STEM graduates to build its own infrastructure, solve its own health problems, and compete globally.

3. Career Pathways in Digital Technology A "digital career" is not just "computer repair." It is a vast field with many specializations.

1. Software Developer (Programmer/Coder):

- **Role:** The "architect" who *designs* and *builds* software. They write the code (in languages like Python, Java, JavaScript) that makes apps, websites, and games work.

2. Cybersecurity Analyst:

- **Role:** The "digital bodyguard." They *protect* a company's computer networks and data from hackers, viruses, and digital theft.

3. **Data Scientist / Data Analyst:**

- **Role:** The "detective." They collect massive amounts of data and *analyze* it to find patterns, make predictions, and help businesses make smart decisions.

4. **UI/UX Designer:**

- **UI (User Interface):** The *look* of an app (the buttons, the colors, the fonts).
- **UX (User Experience):** The *feel* of an app (is it easy to use? Is it confusing? Is it enjoyable?).
- **Role:** They design how the app looks and *feels* to the user.

5. **Network Administrator:**

- **Role:** The "engineer" who *builds* and *manages* a company's internal computer network (the LAN, servers, routers, etc.) to ensure it is fast and reliable.

6. **Digital Marketer:** (From Term 2)

- **Role:** The "promoter" who uses SEO, social media, and content to attract customers online.

7. **Robotics Engineer:**

- **Role:** The "builder" who designs, constructs, and programs robots and automation systems.

4. **Skills Needed for a Digital Career**

• **A. Technical Skills (Hard Skills):**

- You must learn the specific technical skill for your chosen path (e.g., learning to code in Python, learning to use cybersecurity tools, learning to design in Canva/Figma).

• **B. Soft Skills (Essential):**

- These are just as, or even *more*, important than technical skills.

2. **Critical Thinking & Problem Solving:** (The #1 skill)

3. **Continuous Learning:** Technology changes *every day*. Your knowledge from 10 years ago is useless. You must *love* to learn new things.

4. **Collaboration (Teamwork):** No one builds a major app alone. You must be able to work in a team (as learned in Term 2).

5. **Communication:** You must be able to *explain* complex technical ideas to non-technical people (your boss, your client).

Evaluation:

1. What does the acronym "STEM" stand for?
2. Why is STEM education important for Nigeria?
3. List three (3) different career paths in technology.
4. What is the main role of a "Cybersecurity Analyst"?
5. What is the difference between a "UI Designer" and a "UX Designer"?

Assignment:

1. Define STEM.
2. List three "hard skills" and three "soft skills" needed for a digital future.
3. Of the careers listed (Software Developer, Cybersecurity Analyst, Data Scientist, UI/UX Designer, Digital Marketer), which one sounds most interesting to *you*? Do a 5-minute Google search and write two (2) sentences about what that job involves.
4. What is the difference between a "Software Developer" and a "UI/UX Designer"?
5. Why is "continuous learning" so important in a technology career?

WEEK 6: ARTIFICIAL INTELLIGENCE IN EVERYDAY LIFE



Content: (I) Examples: voice assistants, recommendations.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Artificial Intelligence" (AI) in simple terms.
2. Define "Machine Learning" (ML) as the way AI "learns."
3. Identify at least four (4) examples of AI they use in their everyday life.
4. Differentiate between "Narrow AI" (what we have today) and "General AI" (what we see in movies).

Lesson Content:

1. What is Artificial Intelligence (AI)?

- **Definition:** Artificial Intelligence is a branch of computer science focused on building **smart machines** that can perform tasks that *typically* require human intelligence.
- These tasks include:
 - Learning from experience
 - Recognizing patterns
 - Understanding human language
 - Making decisions
 - Solving problems

2. How Does AI "Learn"? (Machine Learning)

- **Definition: Machine Learning (ML)** is a *subset* of AI. It is the main *technique* used to create AI.
- **Process:** Instead of a programmer writing *exact* rules for every situation, they "train" a program (an "model") by feeding it **massive amounts of data**.
- **Example (How AI learns to recognize a cat):**
 1. You don't write rules like "IF it has whiskers AND pointy ears...".
 2. Instead, you show the ML model **1 million pictures**, labeled "Cat" or "Not Cat."
 3. The model *learns for itself* to identify the *patterns* that make a picture a "cat."
 4. After training, it can accurately identify a cat in a new photo it has never seen before.

3. Artificial Intelligence in YOUR Everyday Life AI is not just in science fiction movies; you use it every single day, mostly without noticing.

1. Voice Assistants (Siri, Google Assistant, Alexa):

- When you say, "Hey Google, what is the weather today?", AI is used for:
 - **Natural Language Processing (NLP):** To *understand* your spoken words.
 - **Search:** To *find* the answer.
 - **Speech Synthesis:** To *speak* the answer back to you in a human-sounding voice.

2. Recommendation Engines (The "For You" Page):

- **Netflix / YouTube:** The AI *learns* your viewing habits. It "knows" you watch a lot of action movies and comedy. It then *recommends* new, similar content, (e.g., "Because you watched...").
- **TikTok / Instagram Reels:** The "For You" page is 100% AI. It learns *very* quickly (from your likes, shares, and even how long you watch a video) what kind of content will keep you scrolling.

3. Spam Filters (in Email):

- How does GMail *know* that an email is "spam"?
- It is an AI model trained on *billions* of spam emails. It has learned to recognize the patterns (e.g., words like "winner," "congratulations," "free money," bad spelling) and automatically moves them from your inbox.

4. Navigation Apps (Google Maps / Waze):

- How does Google Maps *know* there is traffic on Third Mainland Bridge *right now*?
- It uses AI to analyze *live, anonymous data* from thousands of other phones (also using Google Maps) that are *moving* slowly on that road. It then *predicts* your arrival time and "re-routes" you.

5. Chatbots (Customer Service):

- When you visit a bank's website and a little window pops up saying, "Hi, I'm [Bank]'s digital assistant! How can I help you?"... that is an AI chatbot.

6. Phone Camera:

- When you take a photo in "Portrait Mode," AI is what identifies the *person* and *blurs* the background.

4. Types of AI

• 1. Narrow AI (or "Weak AI"):

- This is **all the AI we have today**.
- It is an AI that is *specialized* and *trained* to do **one single task** (or a very narrow set of tasks).
- *Example:* The Netflix AI is a *genius* at recommending movies, but it cannot drive a car, write a poem, or diagnose a disease.

- **2. General AI (AGI) (or "Strong AI"):**

- This is the "Hollywood" version of AI (like The Terminator, Ultron, or Data from Star Trek).
- It is a *hypothetical* future AI that would have the *full range* of human intelligence—the ability to reason, plan, learn, and be "conscious" or "self-aware."
- **This does not exist.**

Evaluation:

1. What is Artificial Intelligence (AI)?
2. What is Machine Learning (ML)?
3. What is the main job of a "Recommendation Engine" on YouTube?
4. What is the difference between "Narrow AI" and "General AI"?
5. Which type of AI do we have today? (Narrow AI).

Assignment:

1. Define AI and Machine Learning.
2. List four (4) examples of AI that you (or your family) use every week.
3. Explain how an AI spam filter in your GMail works.
4. What is a "chatbot"?
5. (Opinion) What is one *good* thing AI could do for Nigeria in the future? What is one *bad* thing we should be careful about?

WEEK 8: GLOBAL DIGITAL CITIZENSHIP



Content: (I) Ethics and responsibilities in a connected world.

Performance Objectives: By the end of this lesson, students should be able to:

1. Define "Global Digital Citizenship."
2. Explain why the internet is a "global" community.
3. Describe the responsibility of "Global Awareness."
4. Describe the responsibility of "Cross-Cultural Respect" online.
5. Explain "Digital Empathy" and its role in preventing global conflict.

Lesson Content:

1. What is Global Digital Citizenship?

- **Digital Citizen (Recap):** A person who uses the internet and digital tools safely, responsibly, and ethically.
- **Global Digital Citizenship:** This is the *next level*. It is the understanding that when you go online, you are no longer just a citizen of your *school* or *country* (Nigeria). You are now an active member of a **single, global community** that connects billions of people.
- It is the awareness that your *actions, words, and posts* can be seen by, and can *affect*, anyone, anywhere in the world.

2. Key Responsibilities of a Global Digital Citizen

- **1. Global Awareness & Perspective:**
 - **Responsibility:** To understand that your local problems and views are *not* the only ones.
 - **Action:** Use the internet to *learn* about other cultures, religions, and political situations. Follow news from other countries. Understand that a joke that is "funny" in Nigeria might be deeply offensive to someone in India or Brazil.
- **2. Cross-Cultural Respect (Global Netiquette):**
 - **Responsibility:** To communicate with people from different backgrounds with *respect* and *politeness*, even when you disagree.
 - **Action:** Avoid using stereotypes, racist, or nationalist insults. Do not assume your way is the "only" right way. Be patient with people who are not fluent in your language.
- **3. Digital Empathy:**
 - **Definition:** "Empathy" is the ability to understand and share the feelings of another person. "Digital Empathy" is applying this skill online.
 - **Responsibility:** To *pause* before posting or commenting and ask: "How would this make the *other person* feel?"
 - **Action:** This is the #1 tool to fight cyberbullying (Week 3). Before you post a mean comment, *empathize* with the person in the photo. This prevents online conflict and promotes a healthier global community.

- **4. Global Ethics and Responsibility:**

- **Responsibility:** To understand that your actions contribute to global problems or solutions.
- **Action (Negative):** Participating in international scams ("Yahoo-Yahoo") *damages* the reputation of *all* Nigerians globally. Spreading global misinformation (e.g., about health crises) can cause real-world harm.
- **Action (Positive):** Using the internet to collaborate on global projects (e.g., contributing to Wikipedia, working on an open-source code project, or participating in a global climate change awareness campaign).

3. The "Global Village"

- This is a term used to describe the world, which has been "shrunk" by technology (internet, airplanes).
- A message that once took 6 months to travel by ship now takes 0.6 seconds. We are all "neighbors" in a single "Global Village."
- As a Global Digital Citizen, your responsibility is to be a *good neighbor*.

Evaluation:

1. What is a "Global Digital Citizen"?
2. Why is the internet considered a "global" community?
3. What is "Cross-Cultural Respect"?
4. What is "Digital Empathy"?
5. Explain why "Digital Empathy" is the best way to stop cyberbullying.

Assignment:

1. Define Global Digital Citizenship.
2. What is the "Global Village"?
3. How can participating in an international scam (like "Yahoo-Yahoo") negatively affect other, innocent Nigerians? (Hint: think about global reputation).

4. You are in a global online forum (like Reddit) about football. Someone posts a strong opinion you disagree with, and they use poor English.
 - What is the *bad* (poor citizenship) response?
 - What is the *good* (global digital citizenship) response?
5. What is one thing you can do this week to learn more about another culture using the internet?



WEEK 9: TECHNOLOGY AND SUSTAINABLE DEVELOPMENT



Content: (I) Role of ICT in achieving SDGs.

Performance Objectives: By the end of this lesson, students should be able to:

1. Recall the definition of "Sustainable Development" (from JSS3 Social Studies).
2. Recall the "SDGs" (Sustainable Development Goals).
3. Explain how ICT can help achieve **SDG 3 (Good Health)**.
4. Explain how ICT can help achieve **SDG 4 (Quality Education)**.
5. Explain how ICT can help achieve **SDG 7 (Clean Energy)** and **SDG 8 (Decent Work)**.

Lesson Content:

1. Definitions (Recap)

- **Sustainable Development:** "Development that meets the needs of the present without compromising the ability of future generations to meet their own needs." (Balancing Economy, Society, and Environment).
- **SDGs (Sustainable Development Goals):** The 17 "Global Goals" set by the UN to achieve a better world by 2030 (e.g., No Poverty, Zero Hunger, Quality Education, Climate Action).
- **ICT (Information & Comm. Technology):** The *umbrella term* for all digital technologies (computers, internet, mobile phones, software).

2. The Role of ICT in Achieving the SDGs Technology is one of the most powerful *tools* humanity has to solve the world's biggest problems. ICT can *accelerate* (speed up) our progress toward the SDGs.

- **ICT for SDG 3: Good Health and Well-being**
 - **Telemedicine:** Using video calls (like Zoom or WhatsApp) for "virtual appointments" with a doctor. This allows people in remote villages (who live far from a hospital) to get a diagnosis and medical advice.
 - **Health Information:** Mobile apps (like "mHealth") can track disease outbreaks (like cholera) in real-time, send vaccination reminders to mothers, and give people life-saving health information.
 - **Research:** AI is used to analyze medical data to discover new drugs and treatments faster than ever before.
- **ICT for SDG 4: Quality Education**
 - **Access to Information:** The internet (Google, Wikipedia, YouTube) gives any student with a smartphone access to *more information than is in any physical library*.
 - **E-Learning Platforms:** Websites and apps (like ULesson, Khan Academy, Coursera) provide high-quality video lessons in Math, Science, and Coding, which can be accessed from anywhere.

- **Collaboration Tools:** Google Classroom and Google Docs allow teachers and students to collaborate on assignments digitally.
- **ICT for SDG 7: Affordable and Clean Energy**
 - **Smart Grids (IoT):** Using IoT sensors to build a "smart" electrical grid that monitors energy usage in real-time. This reduces waste, detects faults (like a "NEPA" fault) instantly, and makes it easier to add *renewable* energy (like solar) to the system.
- **ICT for SDG 8: Decent Work and Economic Growth**
 - **Digital Entrepreneurship (Week 1):** The internet and mobile money (Fintech) create a new digital economy, allowing millions of young people to start businesses and create new jobs.
 - **Remote Work:** Collaboration tools (like Zoom and Google Docs) allow people to "work from home," providing decent jobs without needing to move to an overcrowded city.
- **ICT for SDG 16: Peace, Justice, and Strong Institutions**
 - **Fighting Corruption:** Using technology to promote transparency. (e.g., "BudgIT," a Nigerian org, uses ICT to track government spending and publishes it online for all citizens to see).
 - **Digital Identity:** Using secure digital ID (like NIN) can help ensure all citizens have access to legal services and can vote.

Evaluation:

1. What is "Sustainable Development"?
2. What does "SDG" stand for?
3. What is "Telemedicine," and which SDG does it support?
4. How can a student use ICT to achieve SDG 4 (Quality Education)?
5. How does digital entrepreneurship help achieve SDG 8 (Decent Work)?

Assignment:

1. Define "ICT" and "SDGs."

2. Choose one SDG (other than 3, 4, 7, or 8).
3. Go on Google and search: "How can technology help achieve SDG [Your Chosen Goal #]?"
4. Write one paragraph (3-5 sentences) explaining the role of ICT for that goal.
5. What is one *negative* way technology could *hurt* the SDGs? (e.g., e-Waste and SDG 12, or "fake news" and SDG 16).



WEEK 10: FINAL PROJECT: STUDENT DIGITAL INNOVATION



Content: (I) Individual or group project presentation.

Performance Objectives: By the end of this lesson, students should be able to:

1. Identify a local, national, or global problem (e.g., an SDG).
2. Brainstorm a "digital innovation" (a tech-based idea) to help solve it.
3. Use digital tools (Canva, Google Sites, etc.) to create a *prototype* or *presentation* of their idea.
4. Present their idea, problem, and solution to the class.
5. Practice skills in teamwork, critical thinking, and public speaking.

Lesson Content:

1. Project Objective (The "BECE" Project) This is the final, capstone project for all of JSS Digital Technologies. The goal is to act like a *Social Digital Entrepreneur*.

1. **Identify a Problem:** Find a real problem in your school, community, Nigeria, or the world (e.g., any of the SDGs).
2. **Propose a Solution:** Brainstorm a *digital* solution. This could be an app, a website, a social media campaign, or a new tech idea.
3. **Create a Prototype:** *Build* a small "mock-up" of your idea to show the class.
4. **Present (Pitch) Your Idea:** Explain the problem and your solution, like an entrepreneur pitching to an investor.

2. Project Ideas (Group or Individual)

- **Problem:** e-Waste (Week 1)
 - **Solution:** "Green-Tech Initiative."
 - **Prototype:** A 1-page **Google Site** that educates people on e-waste dangers and provides a (mock) map of "drop-off" locations.
- **Problem:** Fake News & Cyberbullying (Term 2)
 - **Solution:** "JSS3 Digital Eagles."
 - **Prototype:** A series of 3 powerful **Canva infographics** (posters) designed for Instagram, each one teaching a "Rule of Netiquette" or "How to Spot Fake News."
- **Problem:** Poor SDG Awareness (Week 9)
 - **Solution:** "SDG Naija Kids."
 - **Prototype:** A simple **Scratch Animation** (from JSS2) or a **PowerPoint presentation** (from Term 1) that explains one of the SDGs (like Clean Water) to JSS1 students.
- **Problem:** Local Business Needs Help (Term 2)
 - **Solution:** "Lagos Student Marketers."

- **Prototype:** A 3-slide **PowerPoint** *pitching* your "Social Media Management" service to a local business, showing a *sample post* you designed for them in Canva.

3. Project Steps (To be done over the week):

1. **Form Groups (Optional).**
2. **Choose a Problem.**
3. **Brainstorm a Solution.**
4. **Assign Tasks:** (e.g., one person on research, one on Canva design, one on Google Site building).
5. **Build the Prototype.**
6. **Practice the Presentation** (3-5 minutes per group).

4. Presentation Structure Each group must present:

1. **The Problem:** "The problem we chose is..."
2. **The Solution:** "Our digital innovation is called... It is a [website/app/campaign]..."
3. **The Prototype:** (Show the class your Google Site, your Canva designs, etc.)
4. **The Impact:** "This will help by..."

Evaluation (Project-Based):

1. (Problem) Was the problem clearly defined and relevant?
2. (Solution) Was the digital solution creative and practical?
3. (Prototype) Was the digital prototype (Canva, Google Site) well-executed and professional?
4. (Presentation) Was the presentation clear, confident, and persuasive?

Assignment:

1. (In class) Form your group and decide on your project topic, solution, and prototype.
2. (At home) Begin work on your prototype.
3. (At home) Write a 1-paragraph "mission statement" for your project.

4. (At home) Prepare your 3-minute presentation script.
5. (Next Week) Be ready to present your final project.



WEEK 11: REVISION



Content: Final review of all years' topics.

Note: This week is dedicated to a **comprehensive review of the entire 3-year JSS Digital Technologies curriculum** in preparation for the Basic Education Certificate Examination (BECE).

Revision Checklist:

- **JSS 1 (Foundations):**
 - **Hardware:** Input (mouse, keyboard), Output (monitor, printer), Storage (HDD, SSD, Flash drive), CPU (the brain).

- **Software:** System (Windows, Android) vs. Application (MS Word, WhatsApp).
- **MS Word/PowerPoint:** Basic document/slide creation.
- **Online Safety (Basic):** What is a password? What is cyberbullying?
- **Computer Ethics (Basic):** Do not steal, do not copy (plagiarism).
- **JSS 2 (Building Blocks):**
 - **Networking:** LAN, MAN, WAN.
 - **Programming:** Algorithm, Debugging.
 - **Block Coding (Scratch):** Sprites, scripts, stage.
 - **Core Concepts:** Variables (storing data), Conditionals (IF/ELSE, decisions), Loops (repetition).
 - **Robotics:** Sensors (senses), Controllers (brain), Actuators (muscles).
 - **AI (Intro):** Meaning of AI.
- **JSS 3 (Real-World Application):**
 - **Term 1 (Digital Media):**
 - Advanced Word (Mail Merge), Advanced PowerPoint (Animation, Transition).
 - Graphic Design (Canva), Branding (Logo, Color, Consistency).
 - Copyright (legal right) vs. Plagiarism (academic cheating).
 - Fair Use vs. Creative Commons (CC).
 - Digital Portfolio (Google Sites).
 - **Term 2 (Digital Entrepreneurship & Ethics):**
 - Digital Business (low cost, global market).
 - Cloud Collaboration (Google Docs).
 - Digital Marketing (SEO, SEM, Content).
 - Netiquette ("Remember the Human," no ALL CAPS).
 - Privacy (PII, Phishing, 2FA).
 - Digital Footprint (Active vs. Passive, permanent reputation).
 - **Term 3 (Emerging Tech & Society):**
 - e-Waste (Lead, Mercury), Safe Disposal (Recycle, Refurbish).
 - Emerging Tech: IoT (smartwatch), Blockchain (secure ledger), AR (filters) vs. VR (headset).

- AI in Daily Life: Recommendations, Voice Assistants, Spam Filter.
- STEM & Digital Careers (Developer, Cybersecurity, Data Scientist).
- Global Citizenship & SDGs.

Activity:

- BECE Past Questions review.
- Final Q&A session.

